BigData Spark Hands-On 5 [part 1]

Spark on k8s
Podman Install

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Outline

Install podman (~Docker) on local Windows/Mac/Linux

Install Kubernetes Cluster "Kind"   (=K8s in Docker)

Install Kubernetes Tools: k9s, helm, aliases

Install spark-operator for K8s using Helm + check/fixes

part 2/2

(Optional) Launch spark-shell on Docker, in mode "--master local[*]"
(Optional) Launch spark with "spark-submit --master k8s://localhost:port"

Launch spark with "kubectl apply -f file.yaml"

(Optional... and backup if local install fails) Using Azure Kubernetes

Pre-Requisites: install K8s

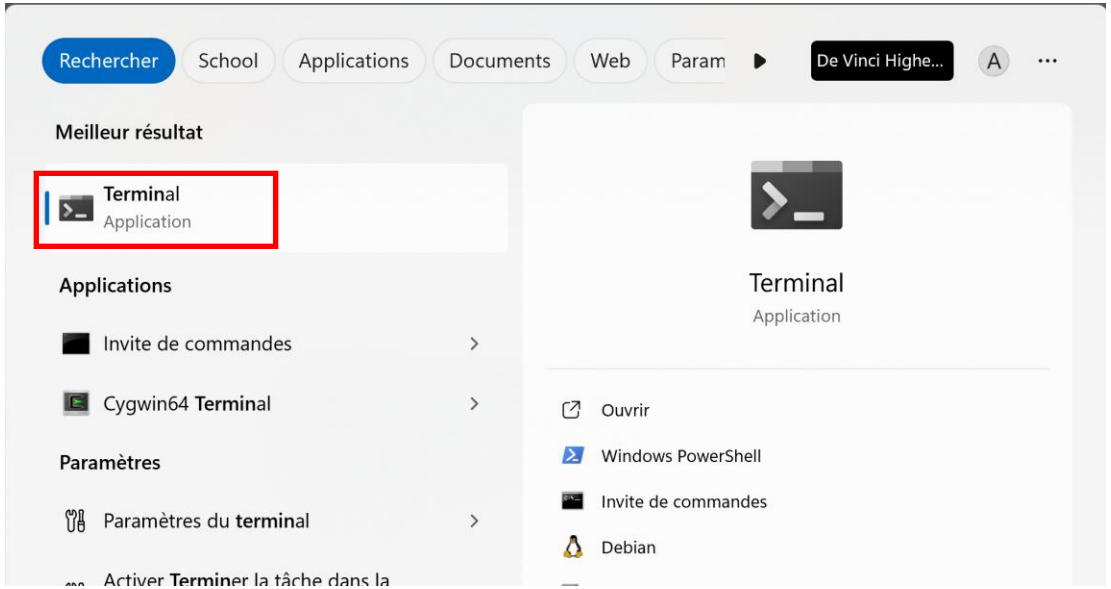
(even on Windows, with podman + Kind)

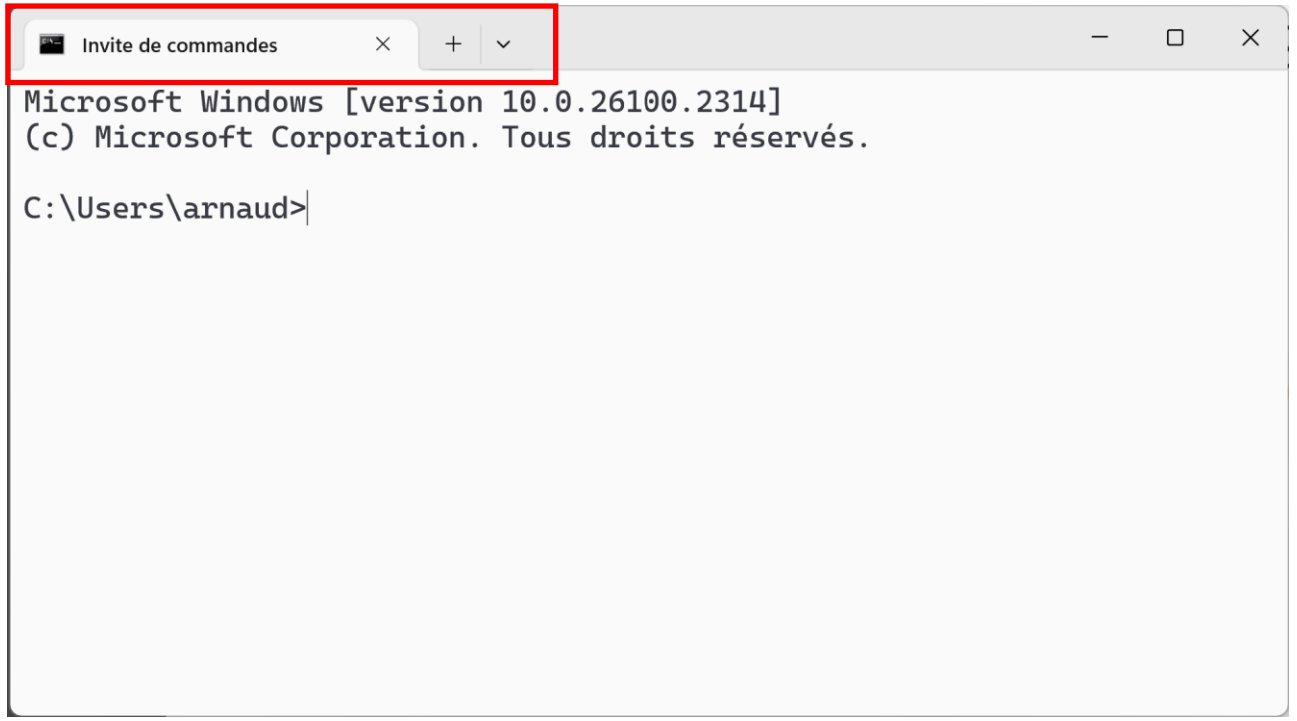
For Windows, Install a modern "Terminal" (do not use "cmd", nor "Power Shell)

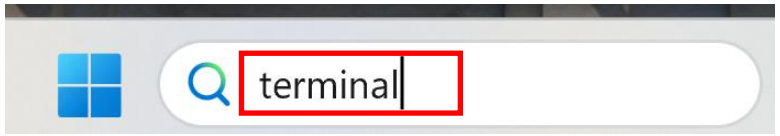
winget install Microsoft.WindowsTerminal

```
C:\apps\spark>winget install Microsoft.WindowsTerminal
Un package existant a déjà été installé. Tentative de mise à niveau du package installé...
Mise à niveau disponible introuvable.
Aucune version de package plus récente n’est disponible à partir des sources configurées.
```

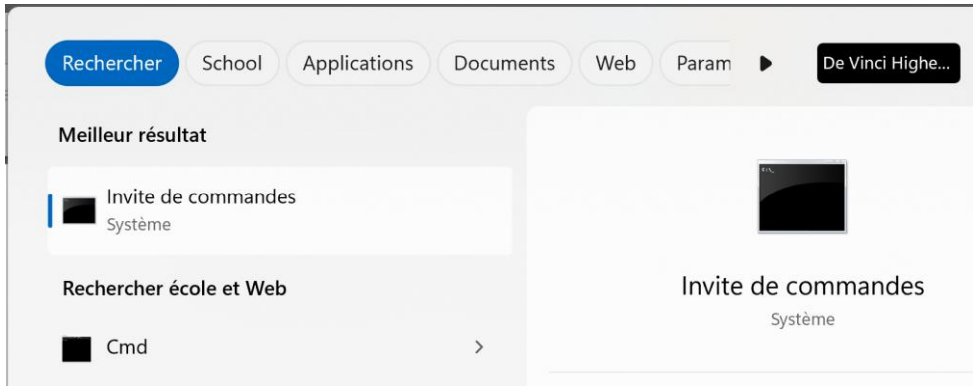
Run Terminal

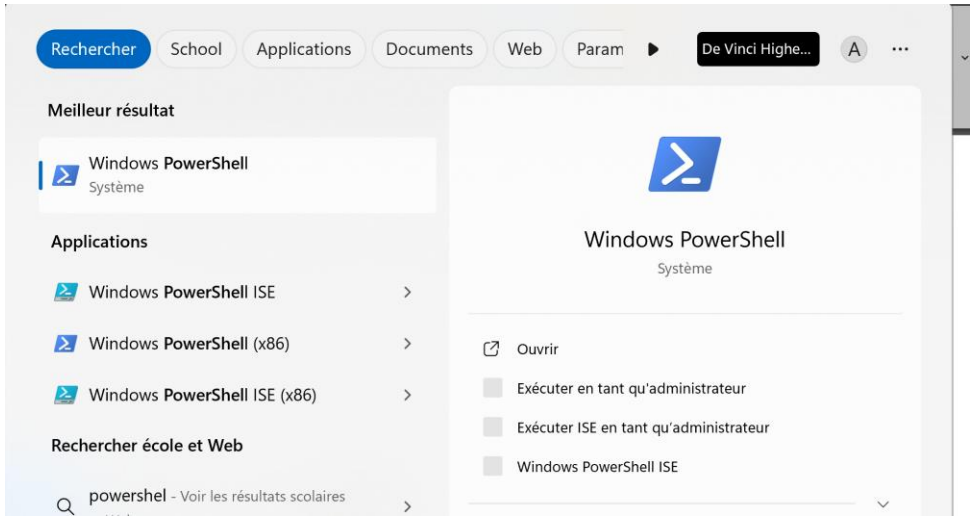


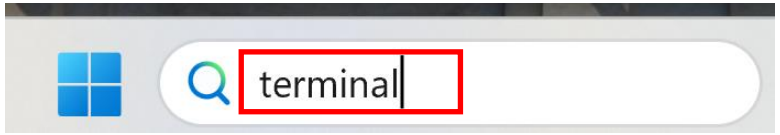


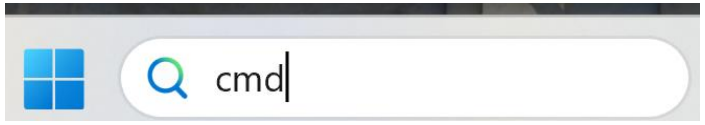


Terminal != Cmd Prompt (Invite de Command), != Power Shell





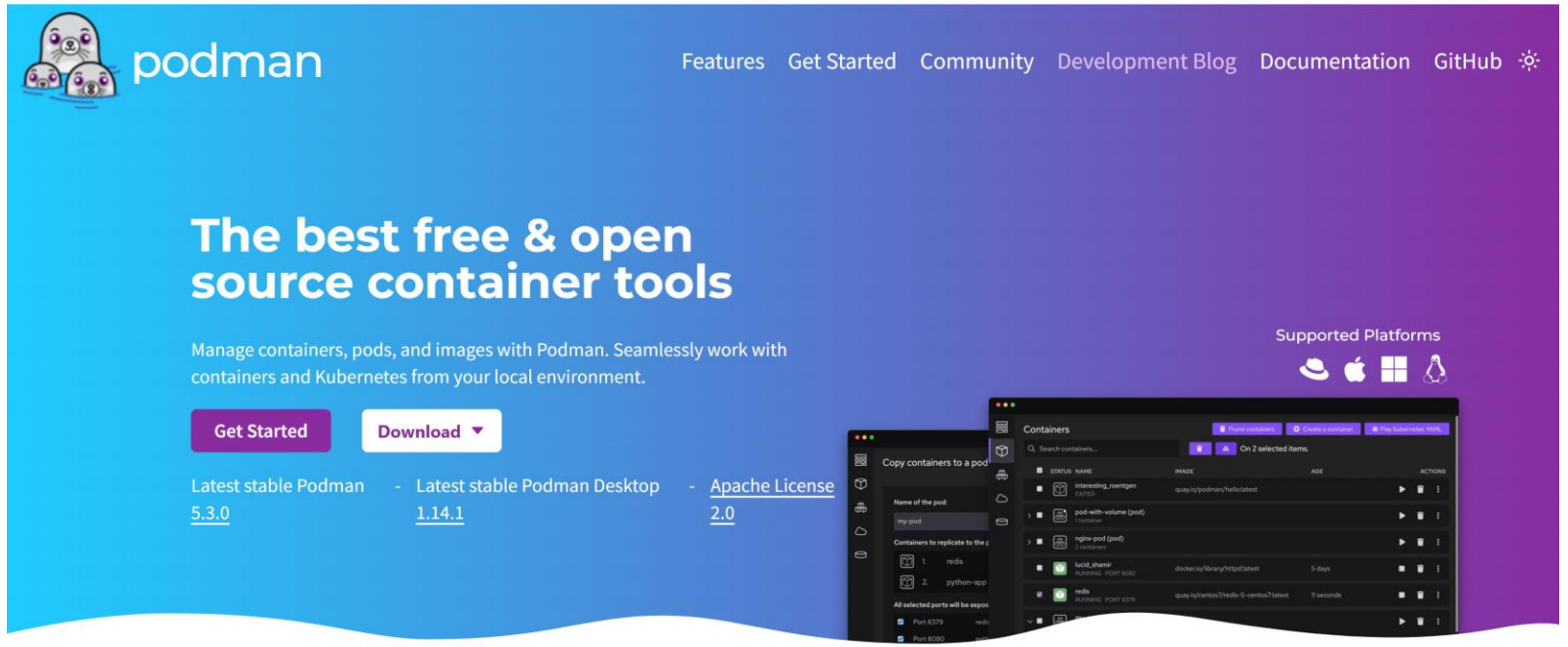






https://podman.io/

Podman is a replacement for "Docker" runtime, for Linux/Windows/Mac/, without root access
Podman also supports tools for easy Kubernetes interaction (minikube / kind / ..)

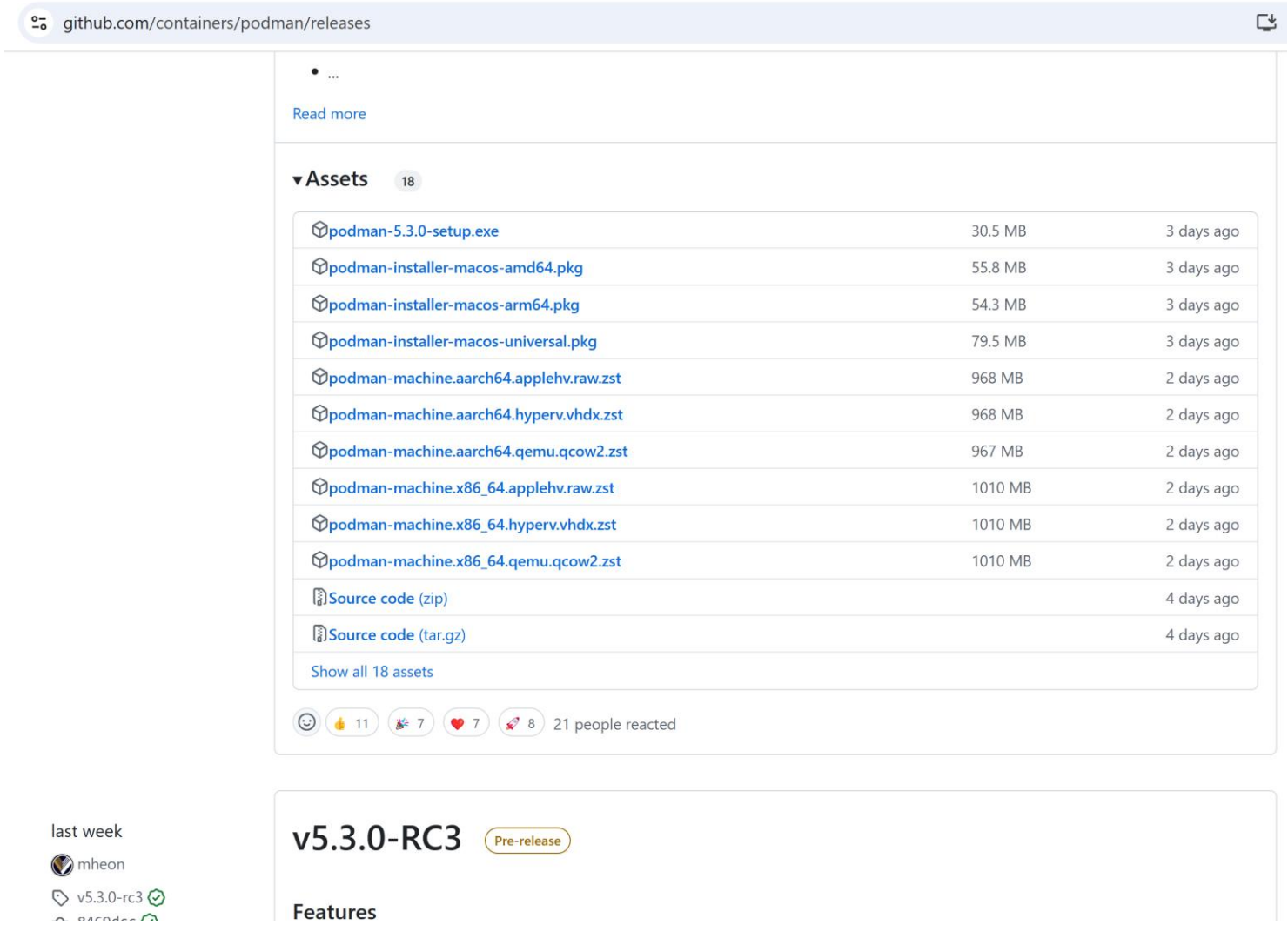


Fast and light.

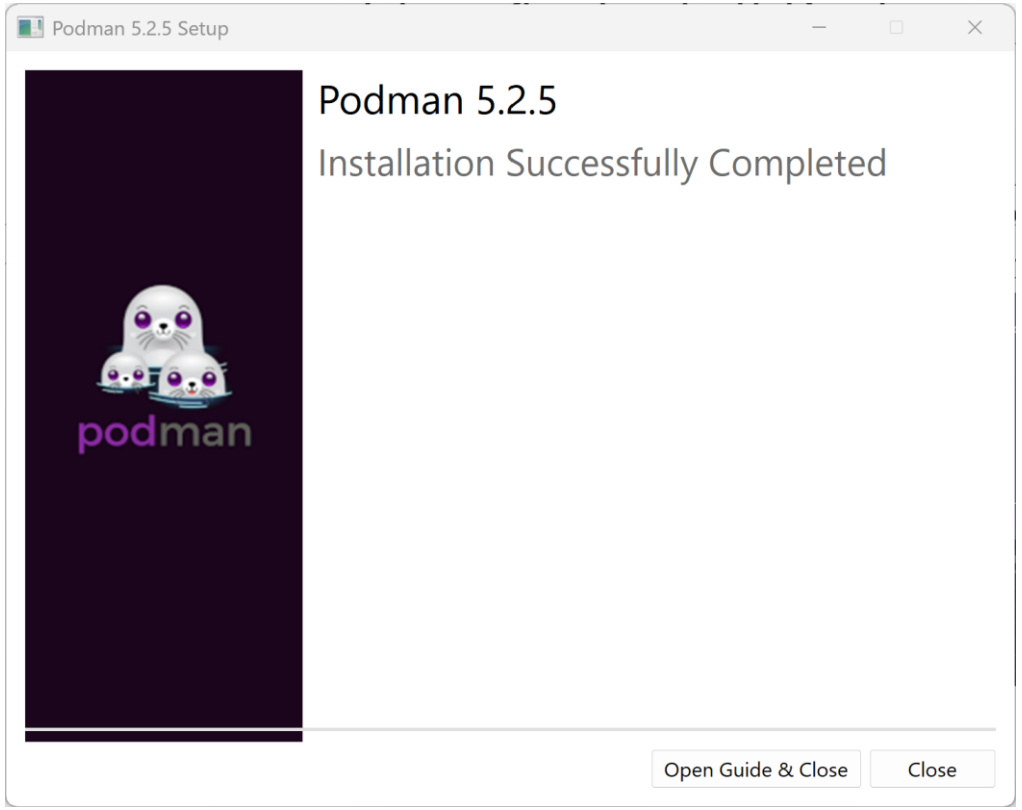
Secure.

Download + exec "podman-{version} .setup.exe"

https://github.com/containers/podman/releases

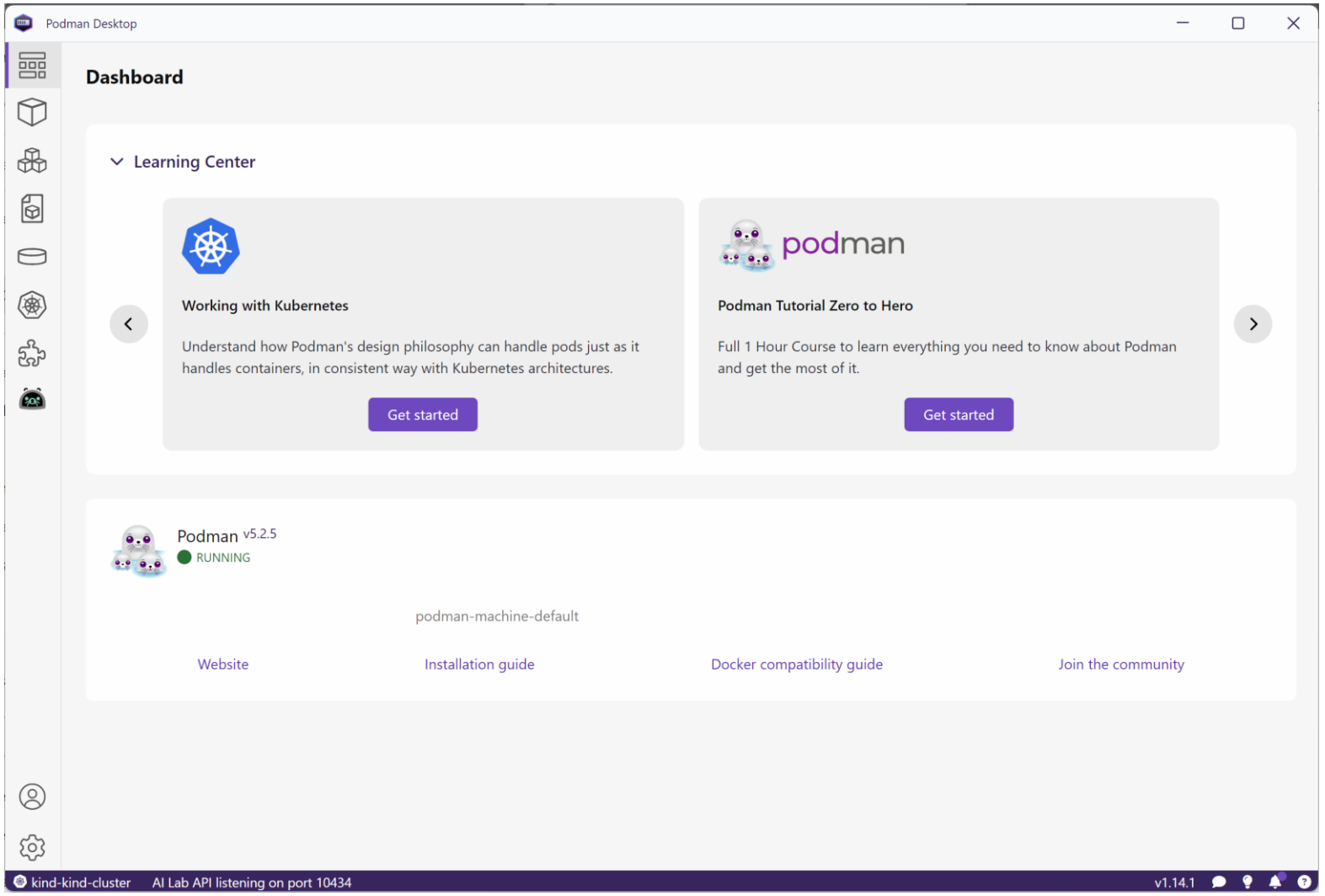


Podman Installed

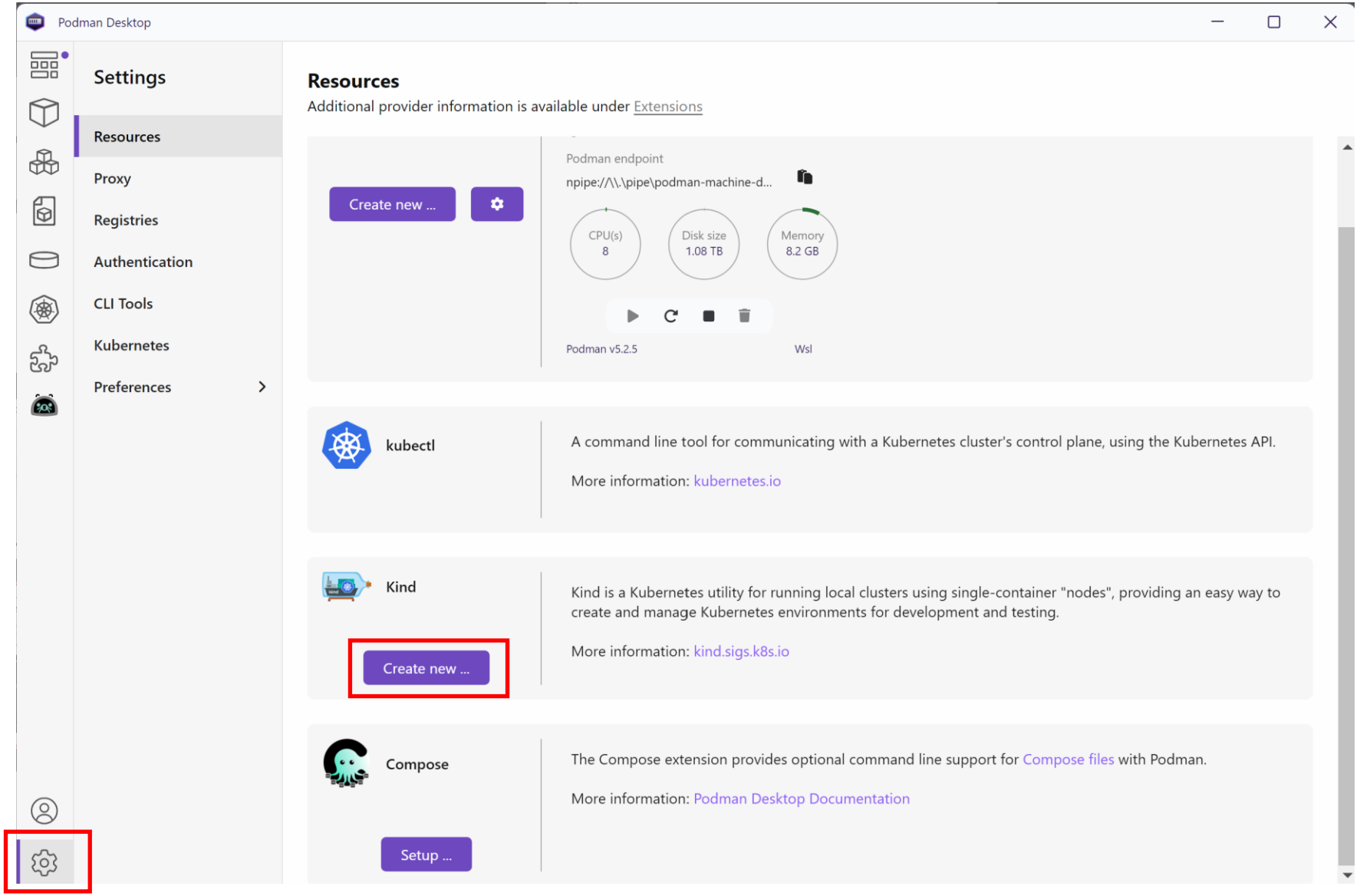


Launch & Explore interface

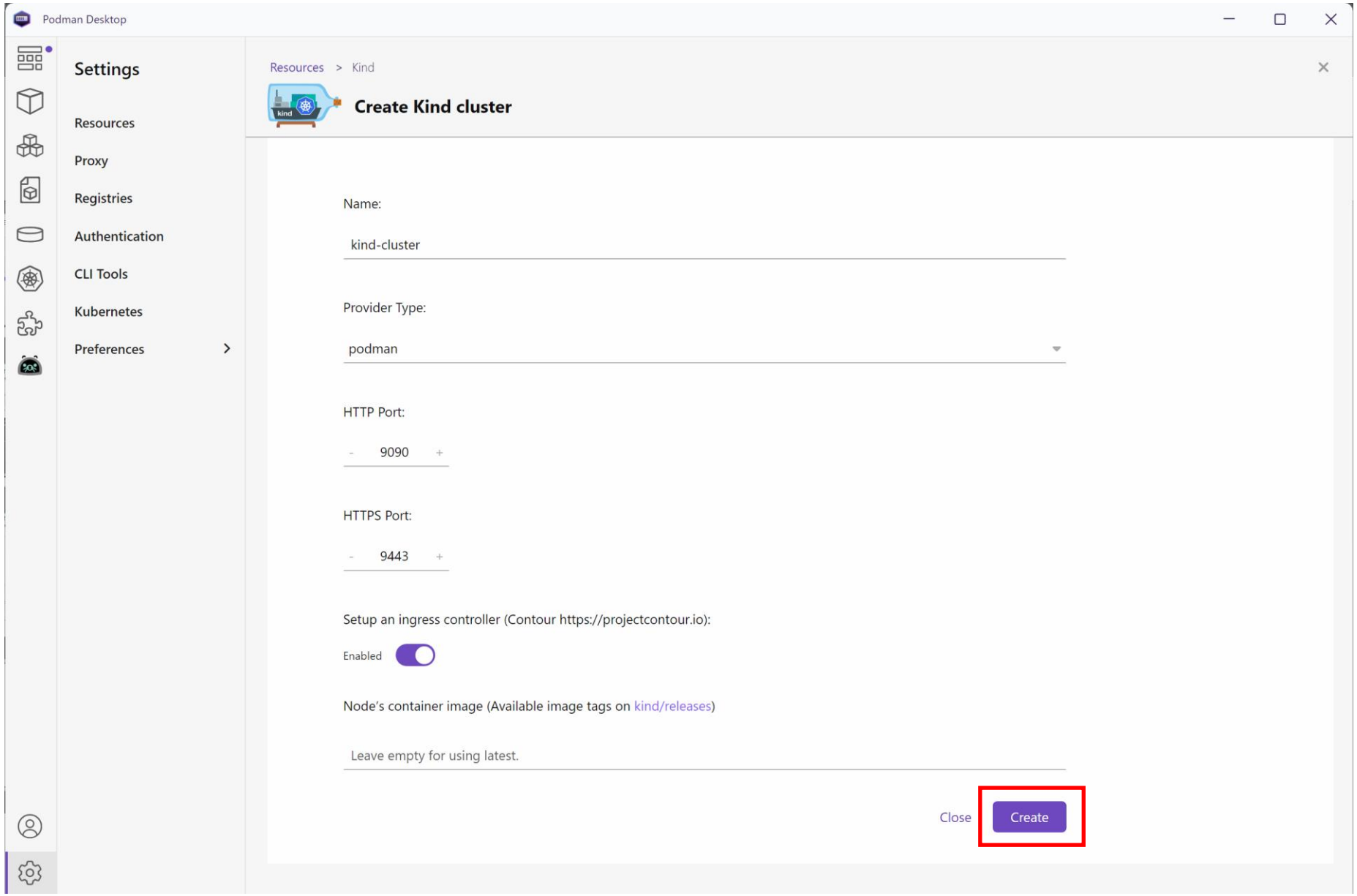




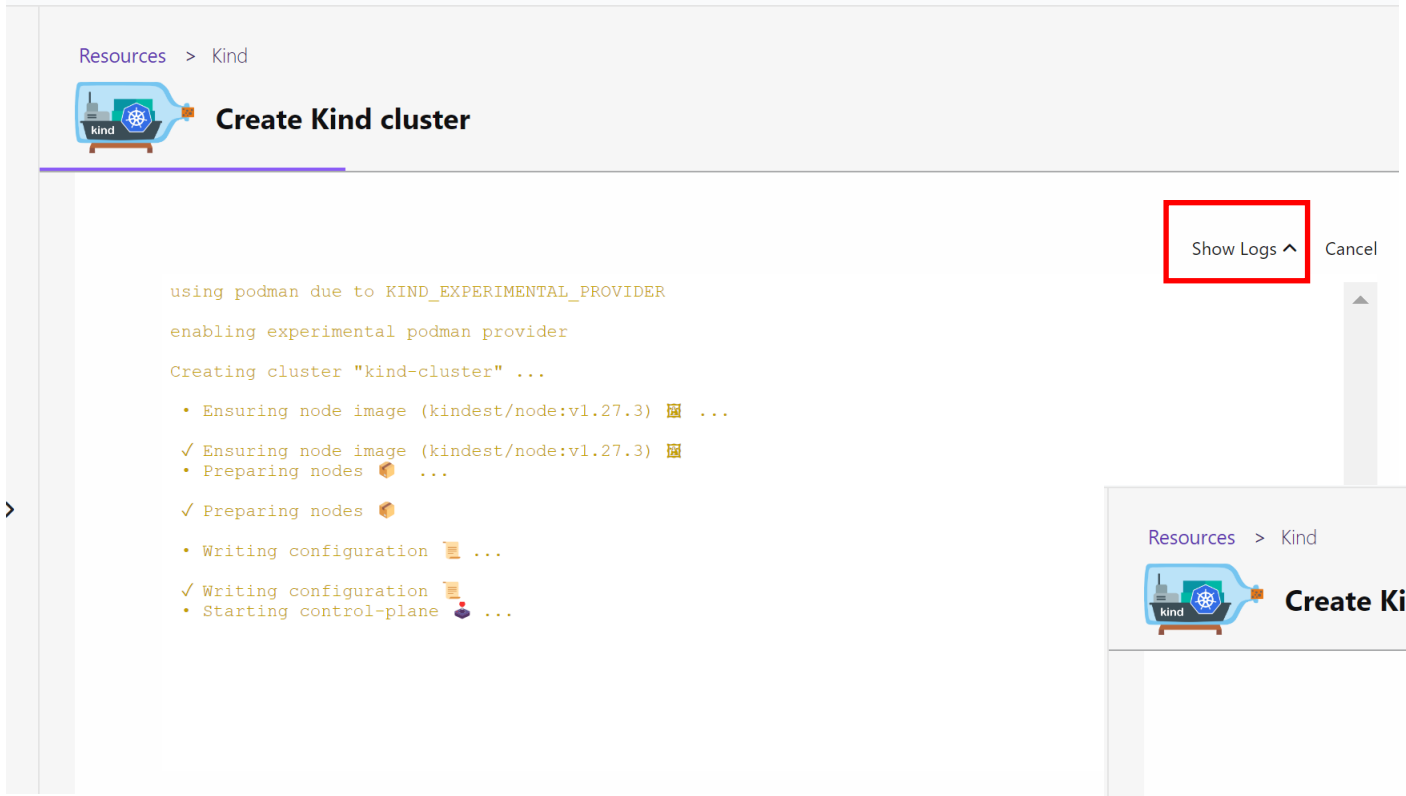
Create a k8s cluster: "Kind" (kind = "K"ubernetes "in" "D"ocker)

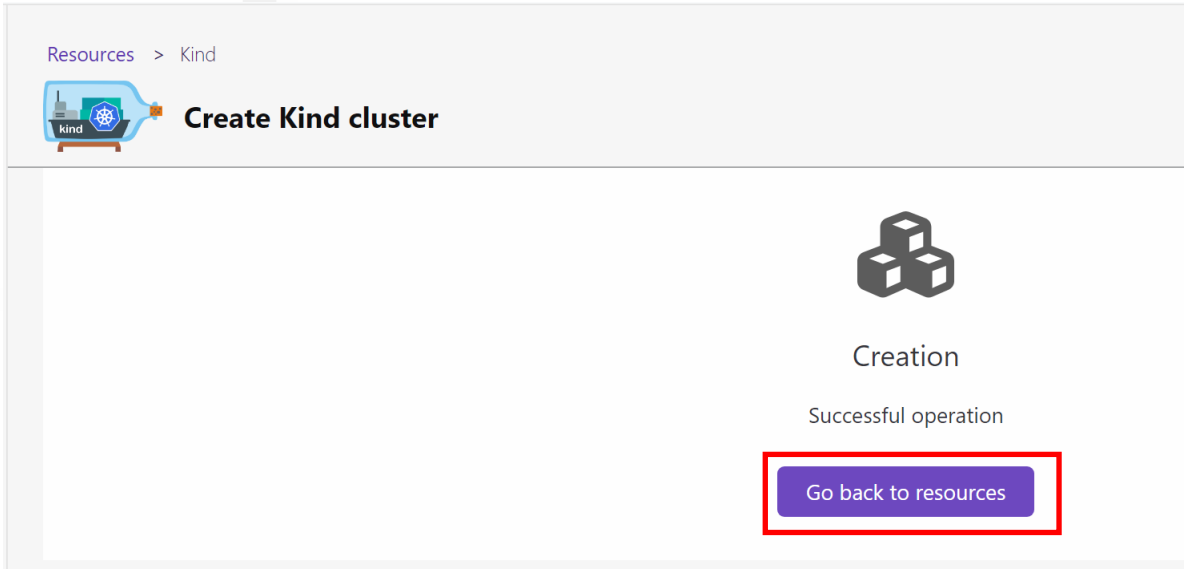


Create Kind

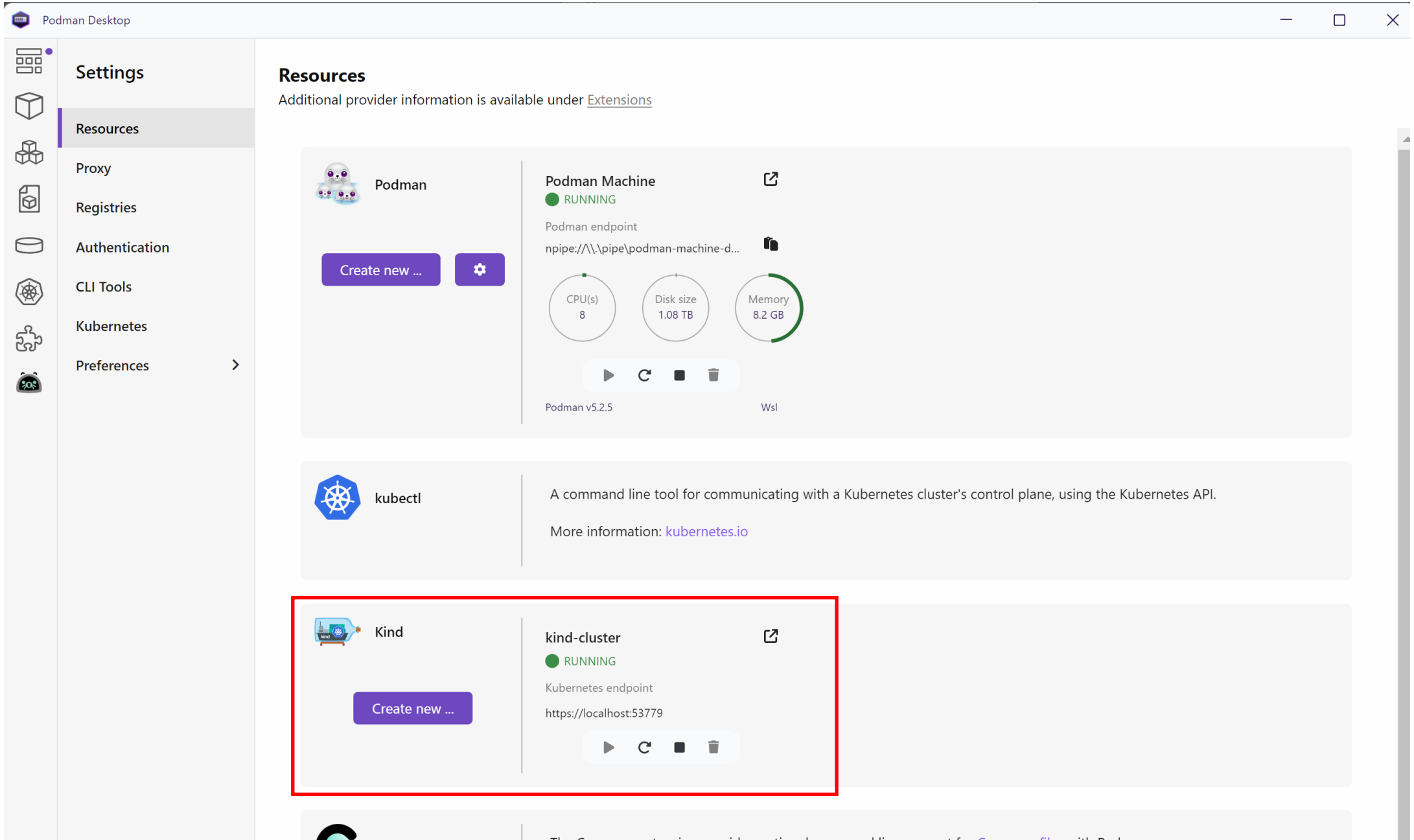

Podman Desktop
Settings
Resources
Proxy
Registries
Authentication
CLI Tools
Kubernetes
Preferences
Resources > Kind
Create Kind cluster
Name:
kind-cluster
Provider Type:
podman
HTTP Port:
- 9090 +
HTTPS Port:
- 9443 +
Setup an ingress controller (Contour https://projectcontour.io):
Enabled
Node's container image (Available image tags on kind/releases)
Leave empty for using latest.
Close
Create

Wait ...

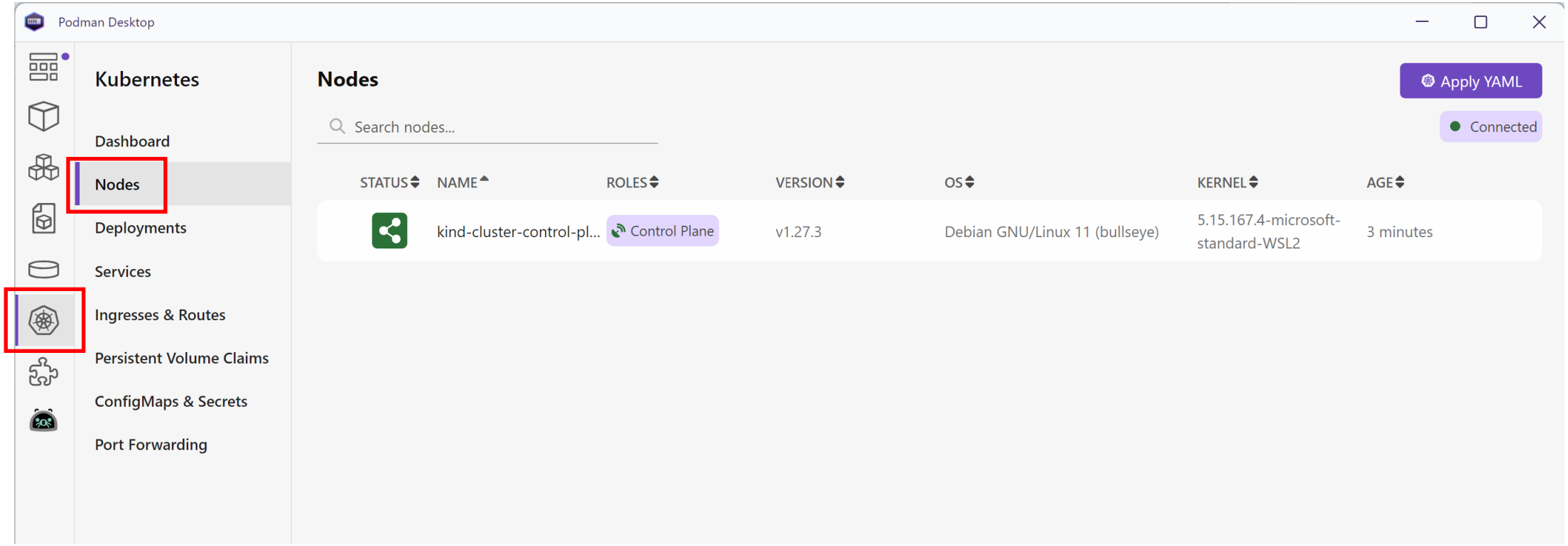

Resources > Kind
Create Kind cluster
Show Logs
Cancel
using podman due to KIND_EXPERIMENTAL_PROVIDER
enabling experimental podman provider
Creating cluster "kind-cluster" ...
• Ensuring node image (kindest/node:v1.27.3) ...
✓ Ensuring node image (kindest/node:v1.27.3)
• Preparing nodes ...
✓ Preparing nodes
• Writing configuration ...
✓ Writing configuration
• Starting control-plane ...


Resources > Kind
Create Kind cluster
Creation
Successful operation
Go back to resources

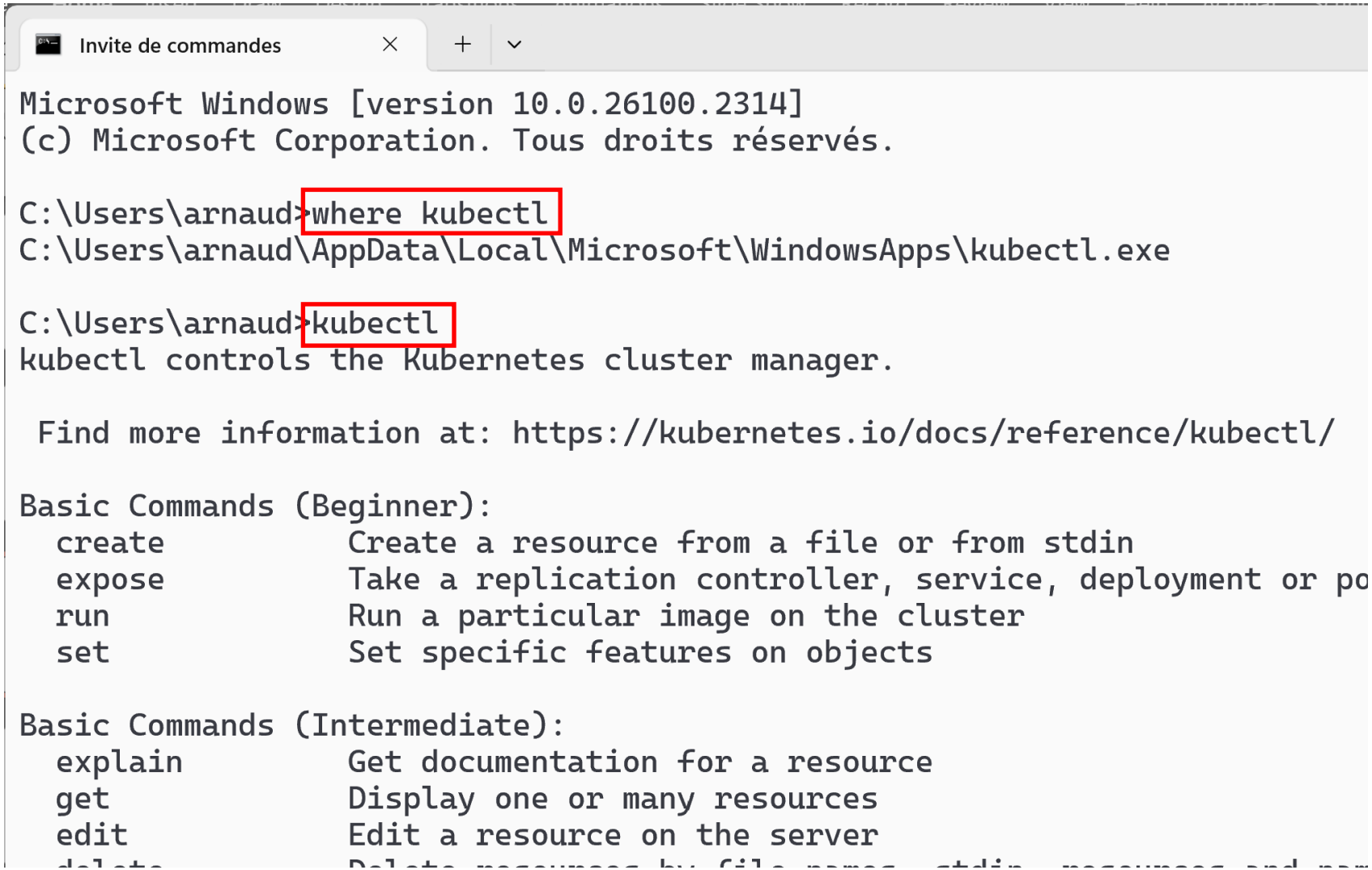
Kind is RUNNING



Kind - Nodes (only 1 : ControlPlane)



Open Terminal, check Kubectl



kubectl config get-contexts
(cf standard aliases next: "kcgc")



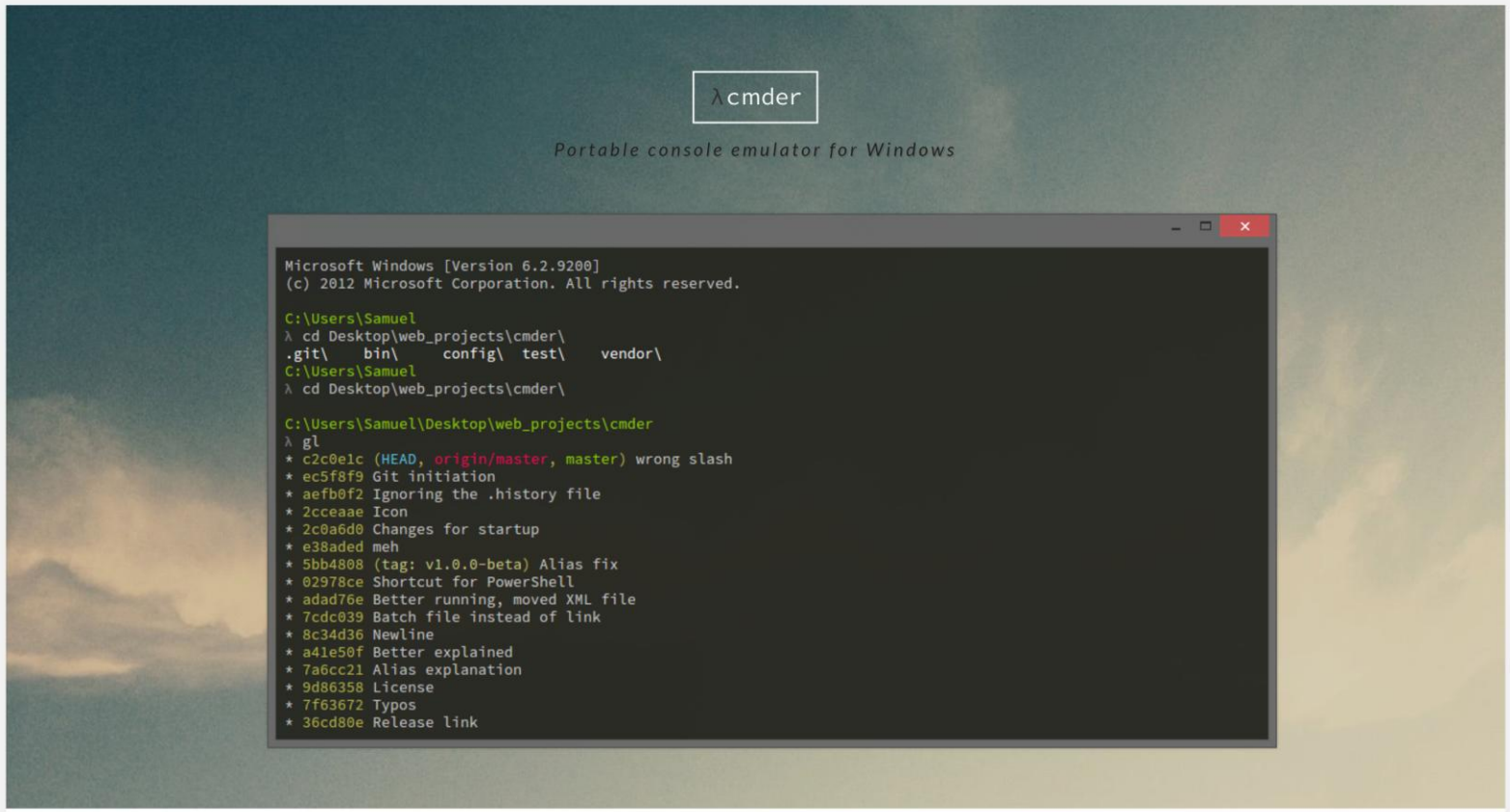
kubectl get ns

(or k get ns … see next for aliases)

```
$ k get ns
NAME                 STATUS   AGE
default              Active   8m47s
kube-node-lease      Active   8m47s
kube-public          Active   8m47s
kube-system          Active   8m47s
local-path-storage   Active   8m36s
projectcontour       Active   8m31s
```

Optional: install K8s helper tools

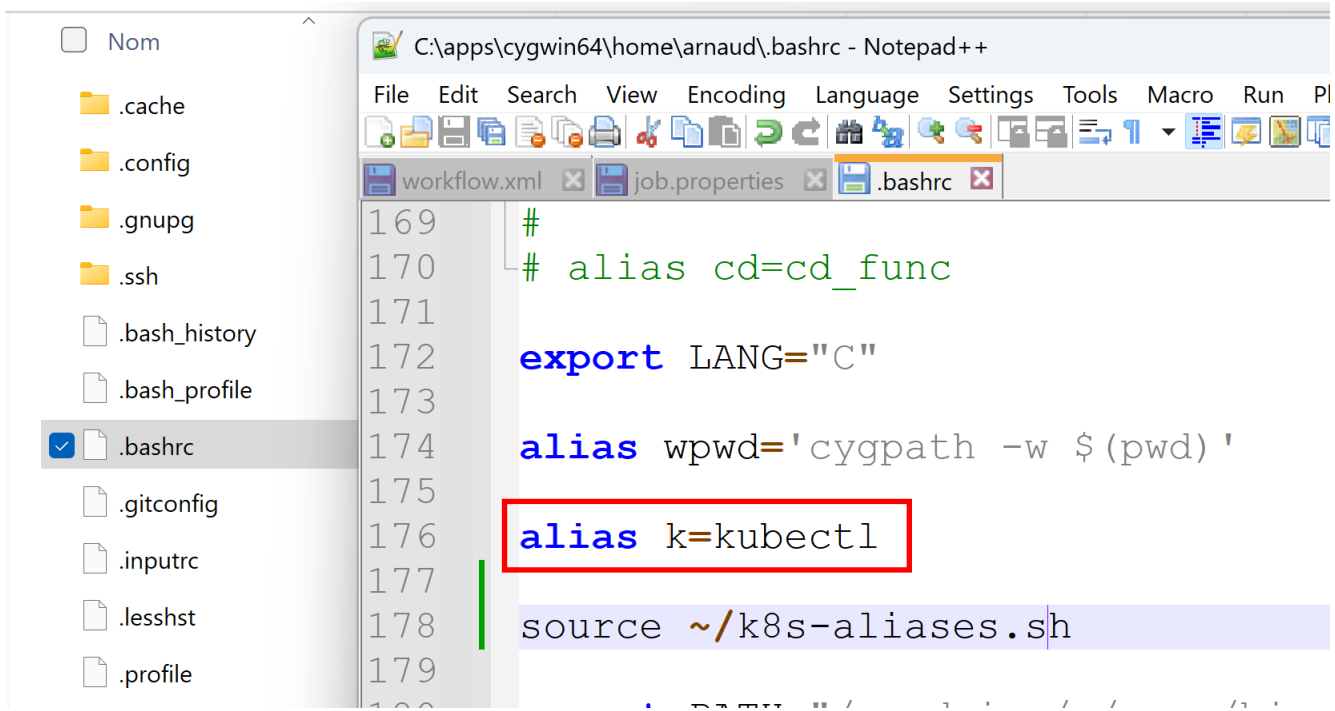
(Optional) Install Cmder https://cmder.app/ for "Bash" (without cygwin) and do not use "cmd DOS" nor "PowerShell" !!



(Optional) usefull on Bash shell : aliases

alias k=kubectl

alias kg=kubectl get; alias kgp=kubectl get pod; ....



(Optional) usefull on Bash shell loading many aliases from "sourced" file

Google

kubectl aliases shell

```
176  alias k=kubectl
177
178  source ~/k8s-aliases.sh
179
```

github.com/dwertent/alias-kubectl/blob/main/kubectl_aliases

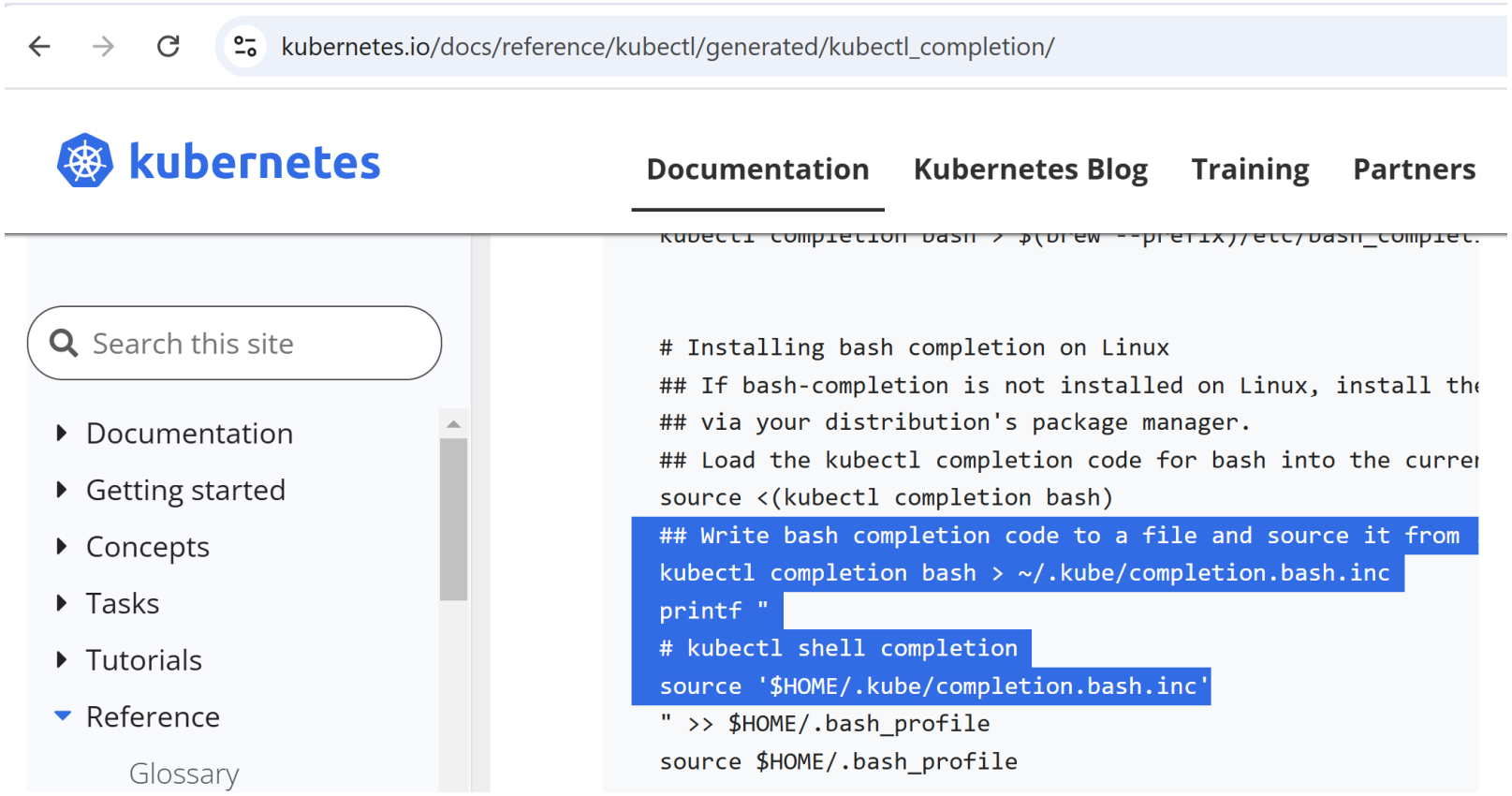
Code | Blame | 120 lines (110 loc) · 3.92 KB | Code 55% faster with GitHub Copilot | Raw

```
#!/bin/bash

# other
alias k='kubectl'
alias kl='kubectl logs'
alias kexec='kubectl exec -it'
alias kpf='kubectl port-forward'
alias kaci='kubectl auth can-i'
alias kat='kubectl attach'
alias kapir='kubectl api-resources'
alias kapiv='kubectl api-versions'

# get
alias kg='kubectl get'
alias kgns='kubectl get ns'
alias kgp='kubectl get pods'
alias kgs='kubectl get secrets'
alias kgd='kubectl get deploy'
alias kgrs='kubectl get rs'
alias kgss='kubectl get sts'
alias kgds='kubectl get ds'
alias kgcm='kubectl get configmap'
alias kgcj='kubectl get cronjob'
alias kgj='kubectl get job'
alias kgsvc='kubectl get svc -o wide'
alias kgn='kubectl get no -o wide'
alias kgr='kubectl get roles'
```

(Optional) Usefull Bash Completions



Kubectl Bash Completions

source <(kubectl completion bash)

```
176  alias k=kubectl
177
178  source ~/k8s-aliases.sh
179
180  # source <(kubectl completion bash)
181  if [ ! -e ~/k8s-completion.sh ]; then
182      kubectl completion bash > ~/k8s-completion.sh;
183  fi
184  source ~/k8s-completion.sh
```

kubectl <cmd><tab> for autocompletion

start typing "ap" then <tab><tab> => propose several auto completions

```
$ kubectl ap
api-resources  (Print the supported API resources on the server)
api-versions   (Print the supported API versions on the server, in the form of "group/version")
apply          (Apply a configuration to a resource by file name or stdin)

arnaud@DesktopArnaud /cygdrive/c/arn/devPerso
$ kubectl ap
```

start typing "app" then <tab> => resolved unique completion "apply"

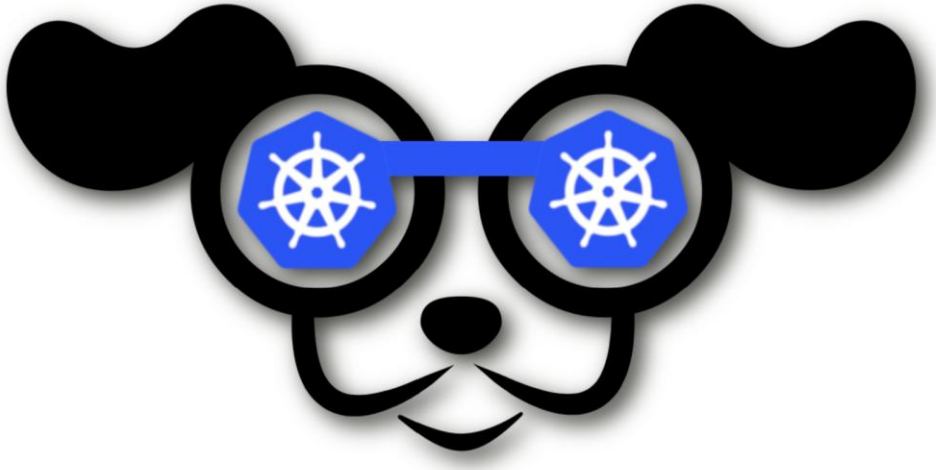
```
$ kubectl app
```

```
$ kubectl apply
```

Install Utility CLI tool : "k9s"

k9s

Kubernetes CLI To Manage Your Clusters In Style!



Who Let The Pods Out?

K9s is a terminal based UI to interact with your Kubernetes clusters. The aim of this project is to make it easier to navigate, observe and manage your deployed applications in the wild. K9s continually watches Kubernetes for changes and offers subsequent commands to interact with your observed resources.

Features

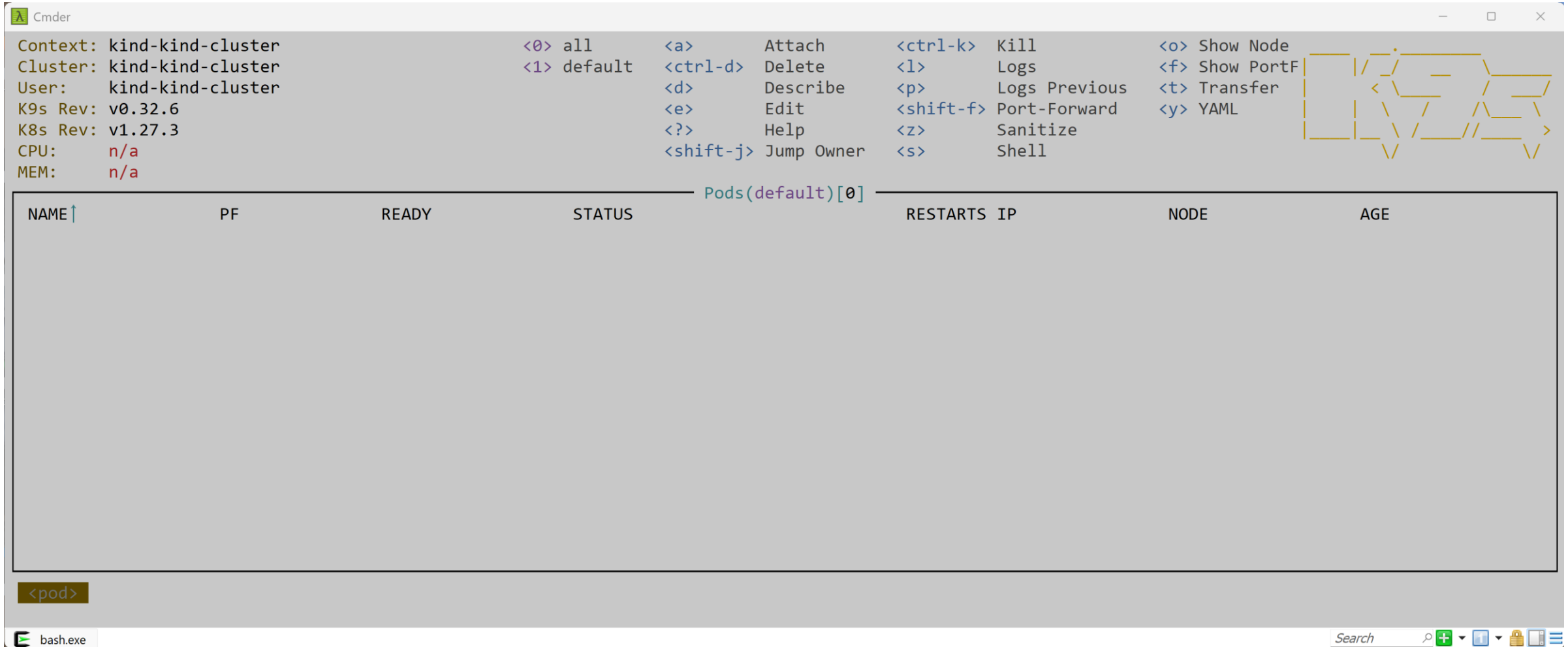
- Information At Your Finger Tips!

https://k9scli.io/topics/install/

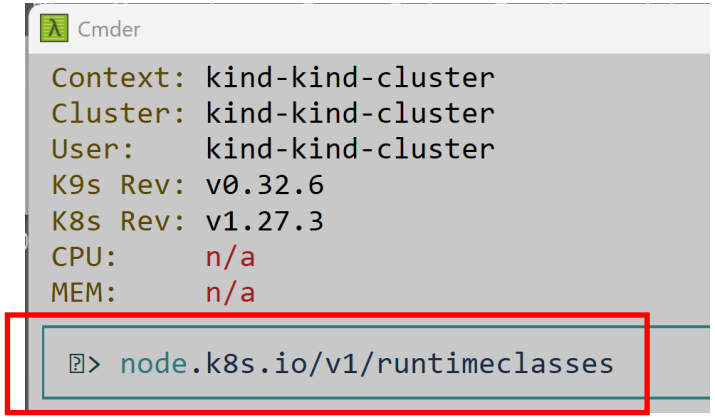
winget install k9s

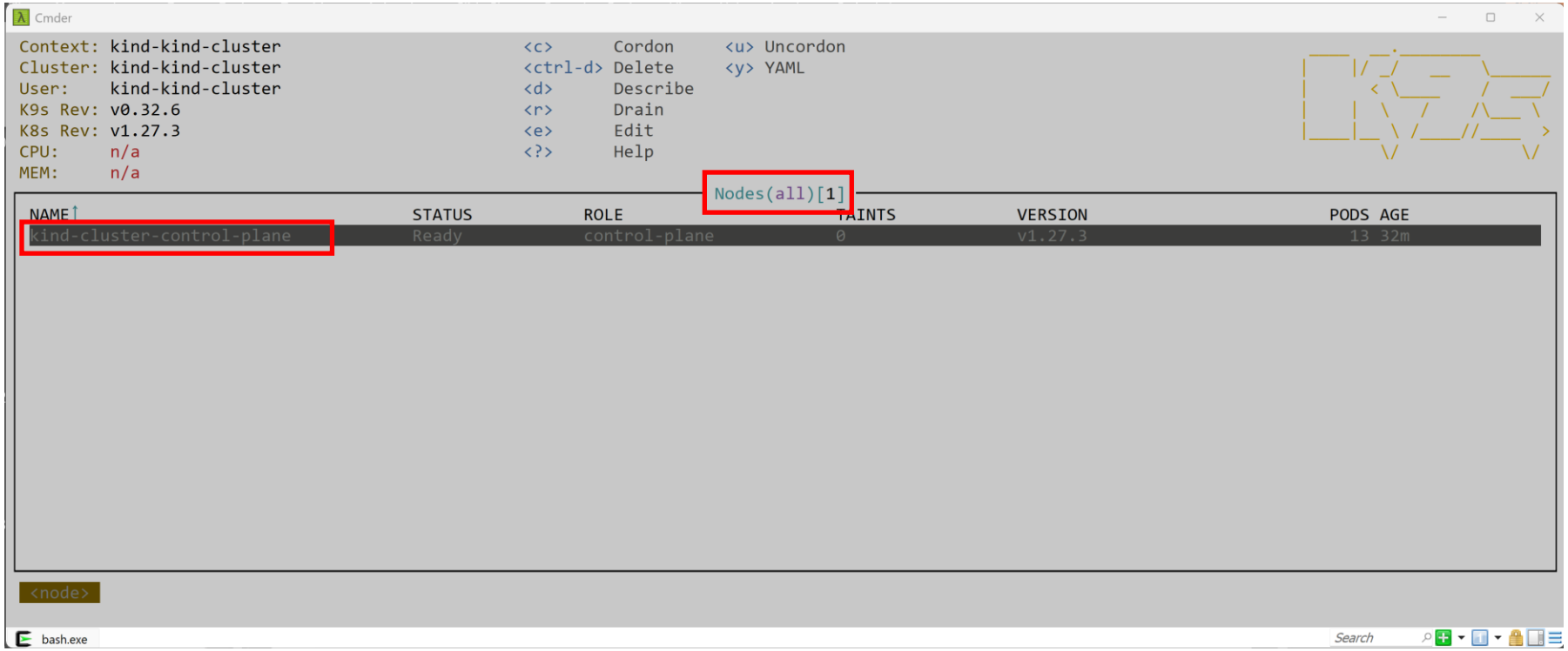
```
$ winget install k9s
Trouvé k9s [Derailed.k9s] Version 0.32.6
La licence d'utilisation de cette application vous est octroyée par son propriétaire.
Microsoft n'est pas responsable des paquets tiers et n'accorde pas de licences à ceux-ci.
Téléchargement en cours https://github.com/derailed/k9s/releases/download/v0.32.6/k9s_Windows_amd64.zip
  ██████████████████████████████  30.6 MB / 30.6 MB
Le code de hachage de l'installation a été vérifié avec succès
Extraction des archives...
Extraction de l'archive réussie
Démarrage du package d'installation... Merci de patienter.
Variable d'environnement de chemin modifiée ; redémarrez votre interpréteur de commandes pour utiliser la nouvelle valeur.
Alias de ligne de commande ajouté : "k9s"
Installé correctement
```

using k9s


Cmder
Context: kind-kind-cluster
Cluster: kind-kind-cluster
User: kind-kind-cluster
K9s Rev: v0.32.6
K8s Rev: v1.27.3
CPU: n/a
MEM: n/a
<0> all
<1> default
<a> Attach
<ctrl-d> Delete
<d> Describe
<e> Edit
<?> Help
<shift-j> Jump Owner
<ctrl-k> Kill
<l> Logs
<p> Logs Previous
<shift-f> Port-Forward
<z> Sanitize
<s> Shell
<o> Show Node
<f> Show PortF
<t> Transfer
<y> YAML
Pods(default)[0]
NAME
PF
READY
STATUS
RESTARTS
IP
NODE
AGE
<pod>
bash.exe
Search

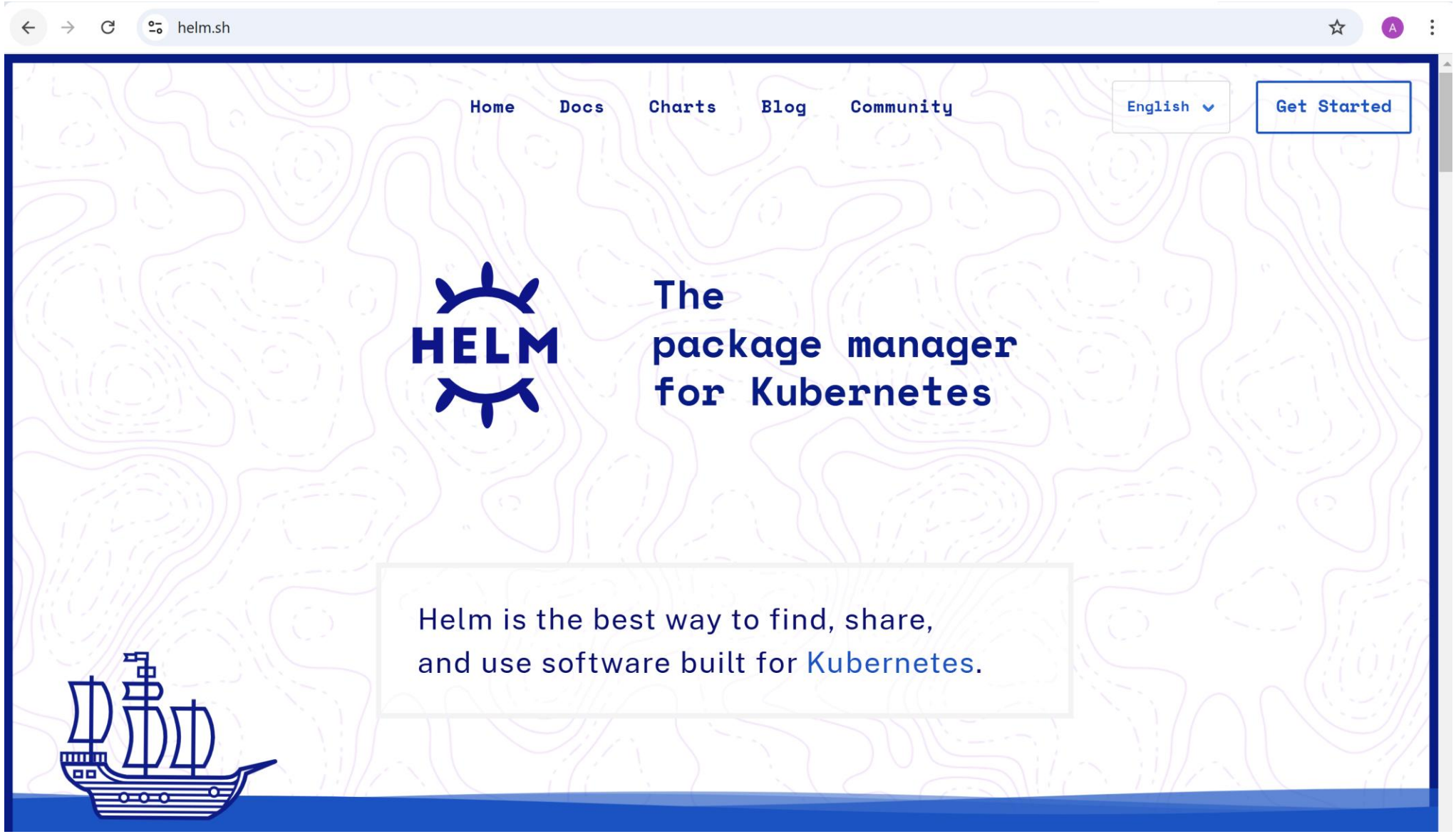
switch between pod, node, deployment: type "<ESCAPE> :node <ENTER>"





Helm

https://helm.sh



Check/Install "Helm" command line

Cmder

```
$ helm
The Kubernetes package manager

Common actions for Helm:

- helm search:    search for charts
- helm pull:      download a chart to your local directory to view
- helm install:   upload the chart to Kubernetes
- helm list:      list releases of charts

Environment variables:

| Name                                | Description
|-------------------------------------|-------------------------------------------------------------
| $HELM_CACHE_HOME                    | set an alternative location for storing cached files.
| $HELM_CONFIG_HOME                   | set an alternative location for storing Helm configuration.
| $HELM_DATA_HOME                     | set an alternative location for storing Helm data.
| $HELM_DEBUG                         | indicate whether or not Helm is running in Debug mode
```

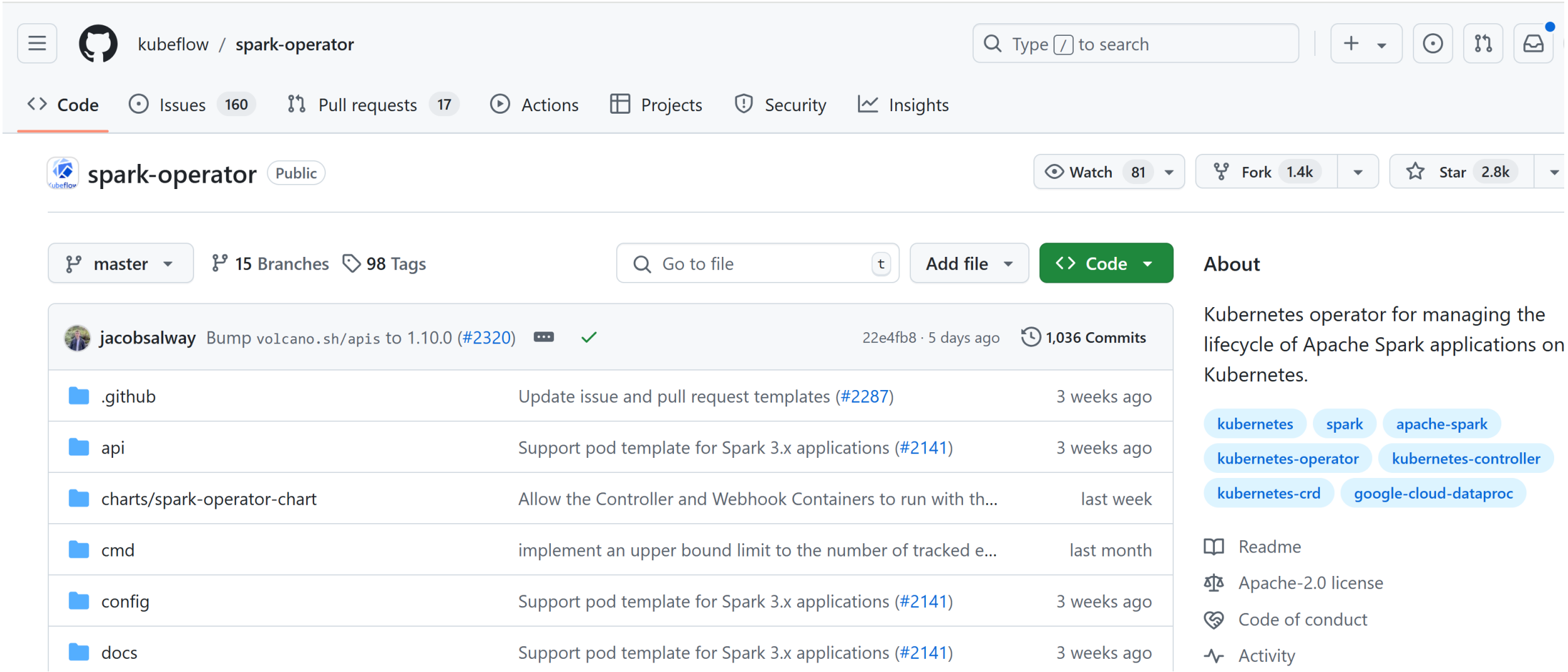
helm list

```
$ helm list
NAME    NAMESPACE       REVISION        UPDATED STATUS  CHART   APP VERSION
```

Nothing installed yet ... We need "spark operator for k8s"

Helm Install
Spark Operator for K8s

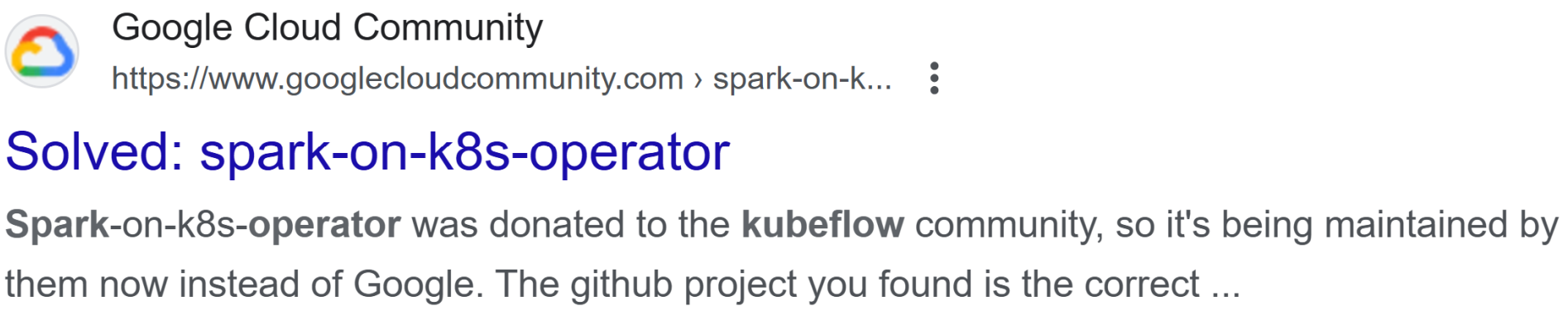
https://github.com/kubeflow/spark-operator



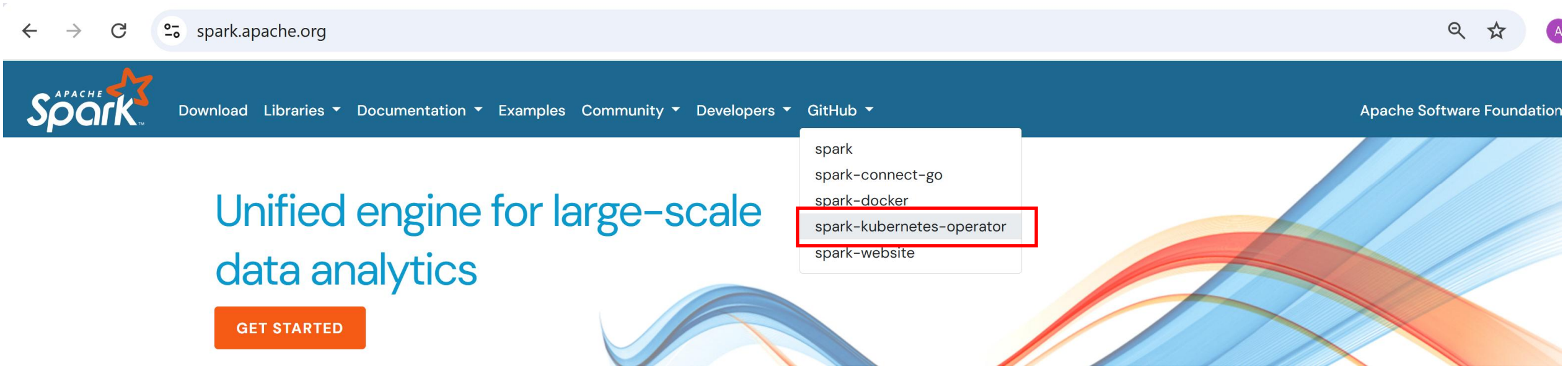
Notice...

https://github.com/GoogleCloudPlatform/spark-on-k8s-operator

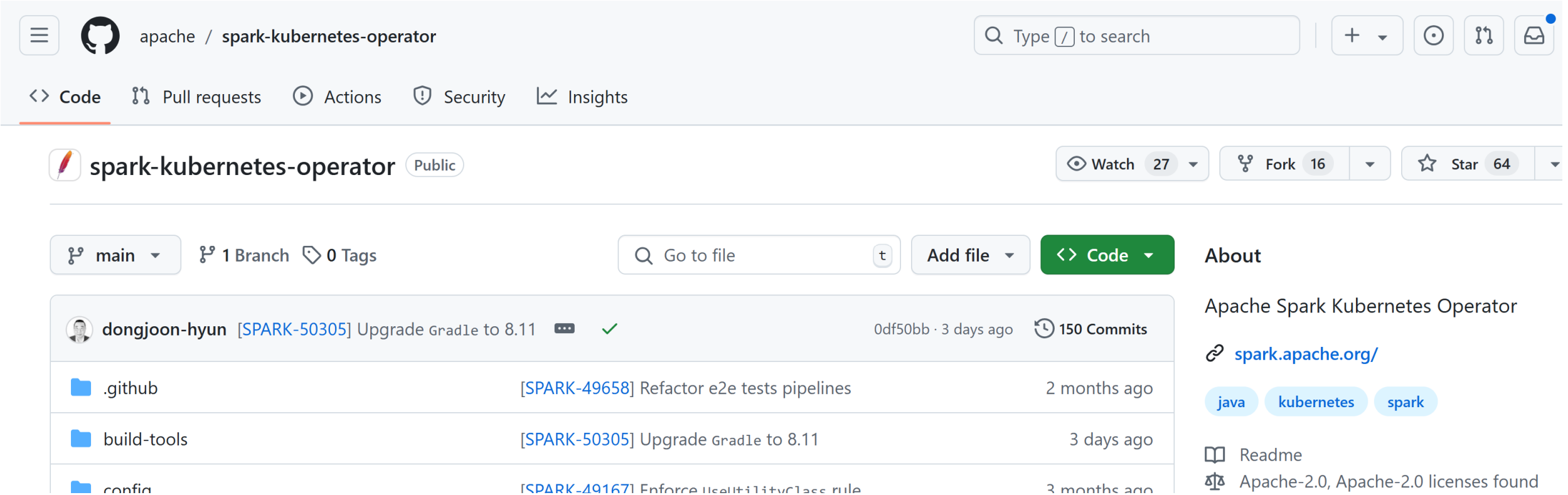
(origin, now automatically redirected to kubeflow)



Notice: another operator…
in apache spark official doc



https://github.com/apache/spark-kubernetes-operator



confusing ...

WARN ...

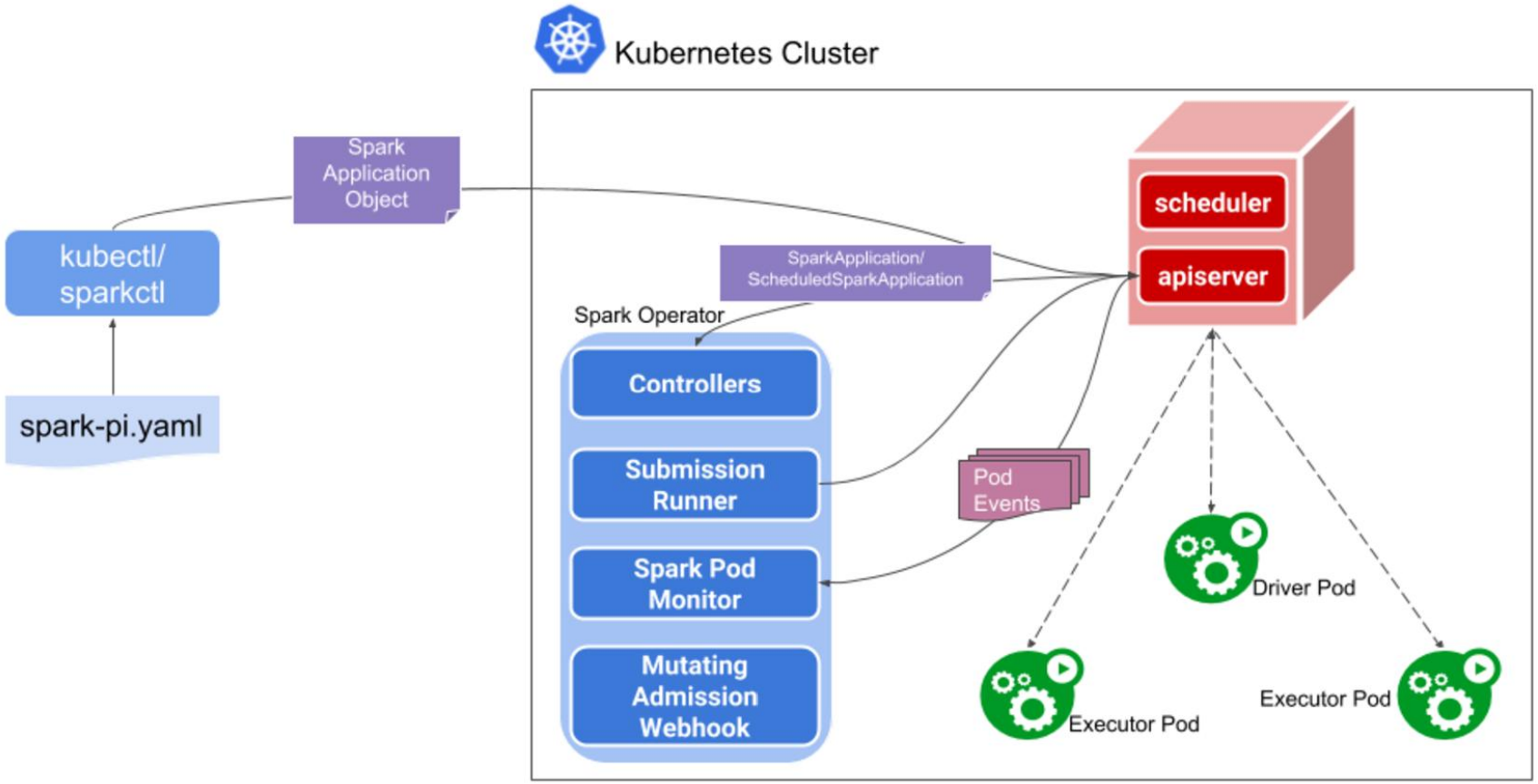
This TD is using the "kubeflow" operator
the kubeflow operator manage these types:

apiVersion: sparkoperator.k8s.io/v1beta2
kind: SparkApplication
kind: ScheduledSparkApplication

!= the "apache" one,
the apache operator manages theses types:

apiVersion: spark.apache.org/v1alpha1
kind: SparkCluster
kind: SparkApplication

doc
https://www.kubeflow.org/docs/components/spark-operator



https://www.kubeflow.org/docs/components/spark-operator/getting-started/

Documentation

- About
- Getting Started
- Concepts
- **Components**
  - Central Dashboard
  - Kubeflow Notebooks
  - Kubeflow Pipelines
  - Katib
  - Model Registry
  - **Spark Operator**
    - Overview

Installation

Add Helm Repo

```
helm repo add spark-operator https://kubeflow.github.io/spark-operator

helm repo update
```

See helm repo for command documentation.

Install the chart

```
helm install [RELEASE_NAME] spark-operator/spark-operator
```

OLD repo already exist (but invalid)
… Failed to update

```
$ helm repo add spark-operator https://kubeflow.github.io/spark-operator
Error: repository name (spark-operator) already exists, please specify a different name

arnaud@DesktopArnaud /cygdrive/c/arn/devPerso
$ helm repo update
Hang tight while we grab the latest from your chart repositories...
...Unable to get an update from the "spark-operator" chart repository (https://googlecloudplatform.github.io/spark-on-k8s-operator):
        failed to fetch https://googlecloudplatform.github.io/spark-on-k8s-operator/index.yaml : 404 Not Found
Update Complete. ⎈Happy Helming!⎈
```

remove + re-add

helm repo **remove** spark-operator
helm repo **add** spark-operator https://kubeflow.github.io/spark-operator
helm repo update

```
$ helm repo remove spark-operator
"spark-operator" has been removed from your repositories

arnaud@DesktopArnaud /cygdrive/c/arn/devPerso
$ helm repo add spark-operator https://kubeflow.github.io/spark-operator
"spark-operator" has been added to your repositories

arnaud@DesktopArnaud /cygdrive/c/arn/devPerso
$ helm repo update
Hang tight while we grab the latest from your chart repositories...
...Successfully got an update from the "spark-operator" chart repository
Update Complete. ⎈Happy Helming!⎈
```

helm install

```
helm install spark-operator spark-operator/spark-operator \
    --namespace spark-operator \
    --create-namespace
```

```
$ helm install spark-operator spark-operator/spark-operator --namespace spark-operator --create-namespace
NAME: spark-operator
LAST DEPLOYED: Sat Nov 16 14:13:45 2024
NAMESPACE: spark-operator
STATUS: deployed
REVISION: 1
TEST SUITE: None
```

Checking: helm list

```
$ helm list
NAME    NAMESPACE       REVISION        UPDATED STATUS  CHART   APP VERSION

arnaud@DesktopArnaud /cygdrive/c/arn/devPerso
$ helm list -n spark-operator
NAME            NAMESPACE       REVISION        UPDATED                                 STATUS          CHART                   APP VERSION
spark-operator  spark-operator  1               2024-11-16 13:22:32.0837027 +0100 CET   deployed        spark-operator-2.0.2    2.0.2
```

NOTICE: helm list => nothing in default namespace, need to specify "-n spark-operator"

installed spark-operator (2.0.2) Not Working ?

checking: kubectl get all -n spark-operator

```
$ kg all -n spark-operator
NAME                                            READY   STATUS                  RESTARTS   AGE
pod/spark-operator-controller-7c47b8c8d7-wnbl9  0/1     CreateContainerError    0          103s
pod/spark-operator-webhook-7f77c86778-9prcj     0/1     CreateContainerError    0          103s

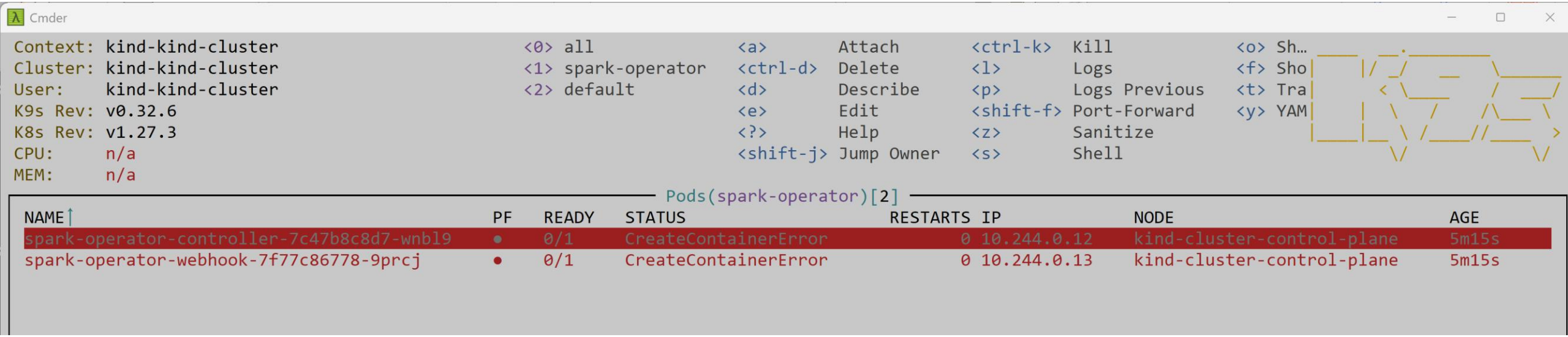
NAME                                  TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)    AGE
service/spark-operator-webhook-svc    ClusterIP   10.96.229.8    <none>        9443/TCP   103s

NAME                                          READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/spark-operator-controller     0/1     1            0           103s
deployment.apps/spark-operator-webhook        0/1     1            0           103s

NAME                                                    DESIRED   CURRENT   READY   AGE
replicaset.apps/spark-operator-controller-7c47b8c8d7    1         1         0       103s
replicaset.apps/spark-operator-webhook-7f77c86778       1         1         0       103s
```

2 errors ... CreateContainerError !

Investigating using k9s



k9s events

```
Context: kind-kind-cluster          <0> all              <d> Describe
Cluster: kind-kind-cluster          <1> spark-operator   <?> Help
User:    kind-kind-cluster          <2> default          <y> YAML
K9s Rev: v0.32.6
K8s Rev: v1.27.3
CPU:     n/a
MEM:     n/a
```

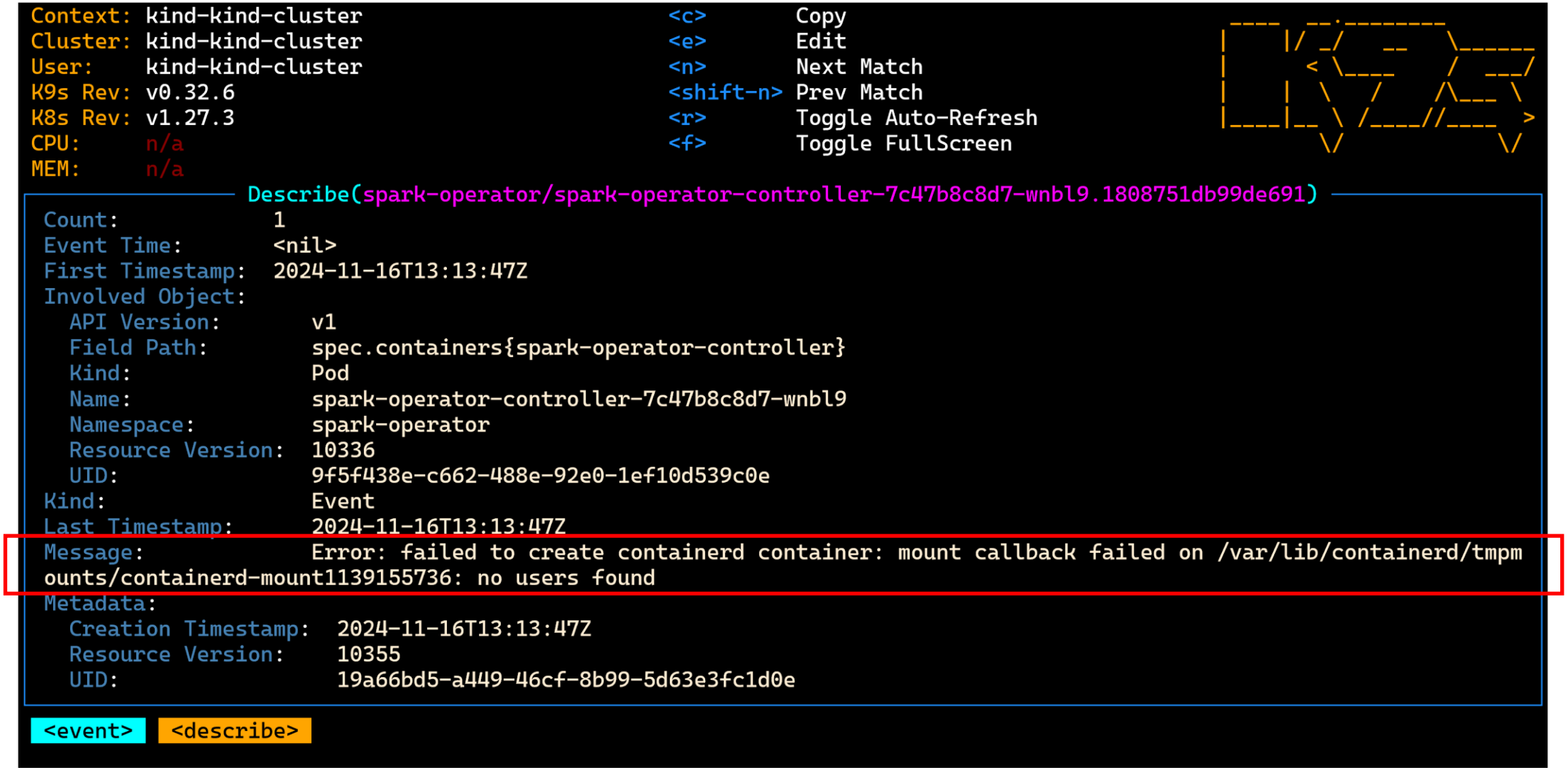


Events(spark-operator)[78]

LAST SEEN↓	TYPE	REASON	OBJECT	COUNT
18m	Normal	SuccessfulCreate	replicaset/spark-operator-controller-7c47b8c8d7	1
18m	Warning	Failed	pod/spark-operator-controller-7c47b8c8d7-wnbl9	1
18m	Warning	Failed	pod/spark-operator-controller-7c47b8c8d7-wnbl9	1
18m	Warning	Failed	pod/spark-operator-controller-7c47b8c8d7-wnbl9	1
18m	Warning	Failed	pod/spark-operator-controller-7c47b8c8d7-wnbl9	1
18m	Warning	Failed	pod/spark-operator-controller-7c47b8c8d7-wnbl9	1
18m	Warning	Failed	pod/spark-operator-controller-7c47b8c8d7-wnbl9	1
18m	Normal	Scheduled	pod/spark-operator-controller-7c47b8c8d7-wnbl9	1
17m	Warning	Failed	pod/spark-operator-webhook-7f77c86778-9prcj	1
17m	Warning	Failed	pod/spark-operator-webhook-7f77c86778-9prcj	1
17m	Warning	Failed	pod/spark-operator-webhook-7f77c86778-9prcj	1
17m	Warning	Failed	pod/spark-operator-controller-7c47b8c8d7-wnbl9	1
17m	Warning	Failed	pod/spark-operator-controller-7c47b8c8d7-wnbl9	1
17m	Warning	Failed	pod/spark-operator-controller-7c47b8c8d7-wnbl9	1
16m	Warning	Failed	pod/spark-operator-webhook-7f77c86778-9prcj	3
16m	Warning	Failed	pod/spark-operator-controller-7c47b8c8d7-wnbl9	3
3m51s	Normal	Pulled	pod/spark-operator-controller-7c47b8c8d7-wnbl9	71
3m49s	Normal	Pulled	pod/spark-operator-webhook-7f77c86778-9prcj	72

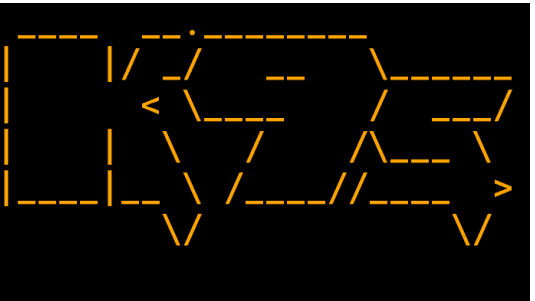
<event>

K9s - event - Failed



k9s - pod - describe

```
Context: kind-kind-cluster                    <c>       Copy
Cluster: kind-kind-cluster                    <e>       Edit
User:    kind-kind-cluster                    <n>       Next Match
K9s Rev: v0.32.6                              <shift-n> Prev Match
K8s Rev: v1.27.3                              <r>       Toggle Auto-Refresh
CPU:     n/a                                  <f>       Toggle FullScreen
MEM:     n/a
```



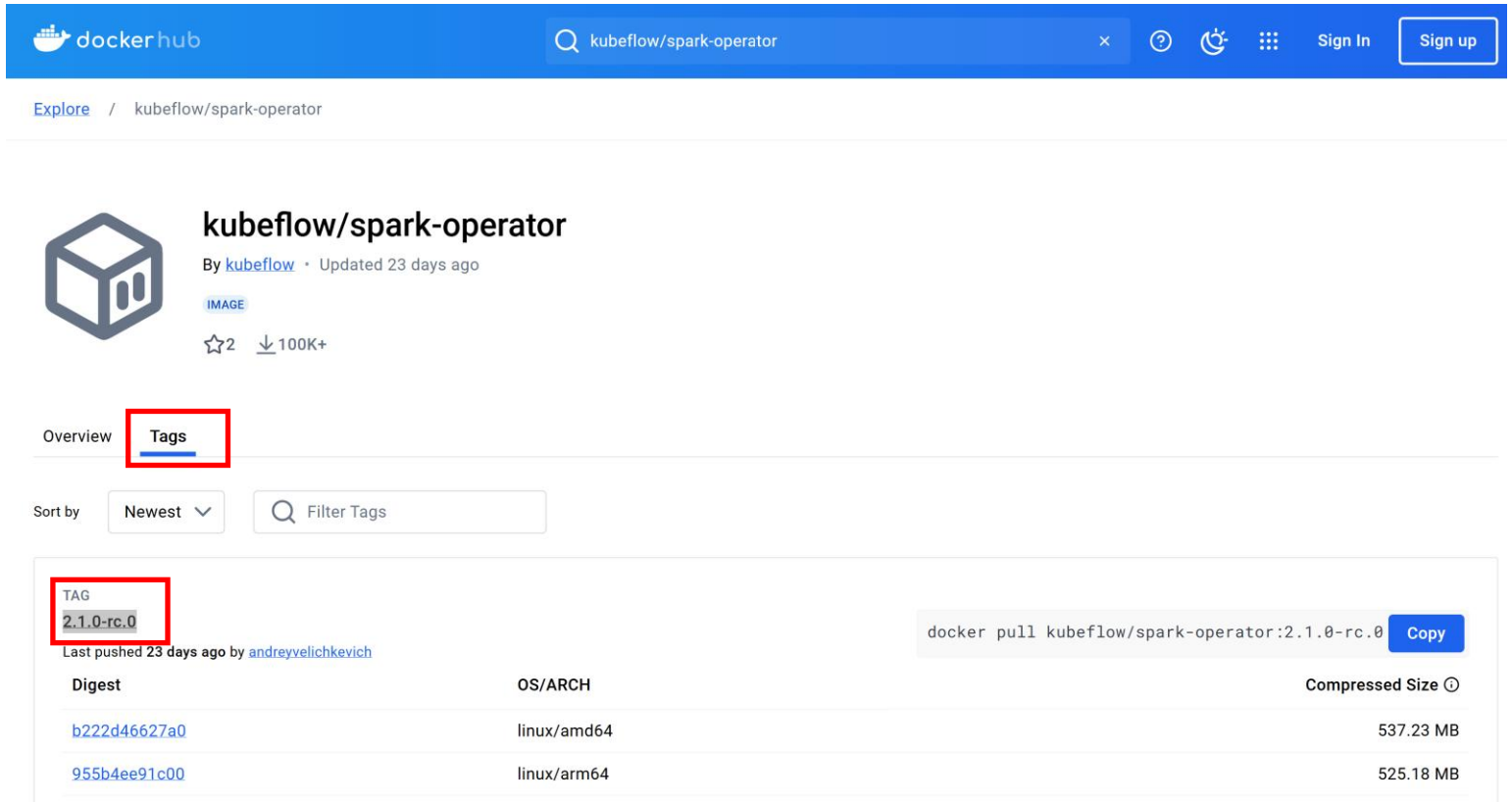
Describe(spark-operator/spark-operator-webhook-7f77c86778-9prcj)

```
allback failed on /var/lib/containerd/tmpmounts/containerd-mount37773448: no users found
  Warning  Failed     16m                    kubelet            Error: failed to create containerd container: mount c
allback failed on /var/lib/containerd/tmpmounts/containerd-mount128027710: no users found
  Warning  Failed     16m                    kubelet            Error: failed to create containerd container: mount c
allback failed on /var/lib/containerd/tmpmounts/containerd-mount2920509095: no users found
  Warning  Failed     16m                    kubelet            Error: failed to create containerd container: mount c
allback failed on /var/lib/containerd/tmpmounts/containerd-mount3338808262: no users found
  Warning  Failed     16m                    kubelet            Error: failed to create containerd container: mount c
allback failed on /var/lib/containerd/tmpmounts/containerd-mount1761742618: no users found
  Warning  Failed     16m                    kubelet            Error: failed to create containerd container: mount c
allback failed on /var/lib/containerd/tmpmounts/containerd-mount2507996816: no users found
  Warning  Failed     15m                    kubelet            Error: failed to create containerd container: mount c
allback failed on /var/lib/containerd/tmpmounts/containerd-mount1222801735: no users found
  Warning  Failed     15m (x3 over 15m)      kubelet            (combined from similar events): Error: failed to crea
te containerd container: mount callback failed on /var/lib/containerd/tmpmounts/containerd-mount3396310179: no users
 found
  Normal   Pulled     2m17s (x72 over 17m)   kubelet            Container image "docker.io/kubeflow/spark-operator:2.
0.2" already present on machine
```

<pod> <describe>

Find latest image

https://hub.docker.com/r/kubeflow/spark-operator/tags



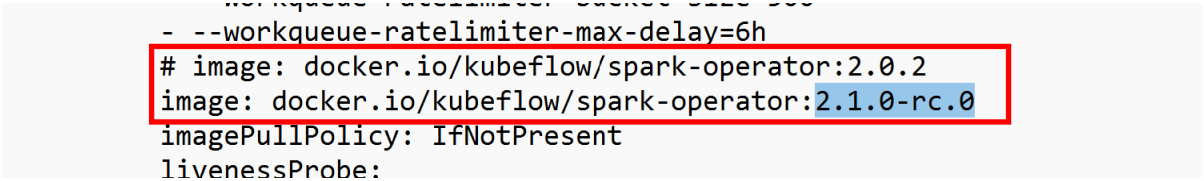
k9s deployment - edit

in K9, type

(if not already set, to select namespace :
<ESCAPE> :ns<ENTER> <ESCAPE> /spark-operator <ENTER>)

<ESCAPE> :deployment <ENTER>
scroll to "spark-operator-controller" (or <ESCAPE> / *controller)
type "e" to Edit

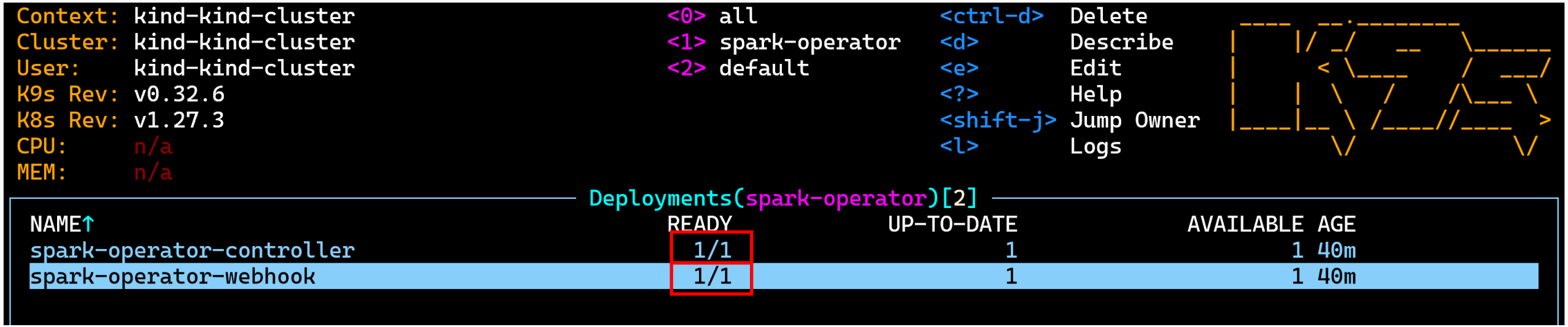
=> your Notepad editor opens, change "2.0.2" to "latest" (or "**2.1.0-rc.0**" as of 2024-11)



then "SAVE" and close editor

... REPEAT same edit for other: "spark-operator-webhook"

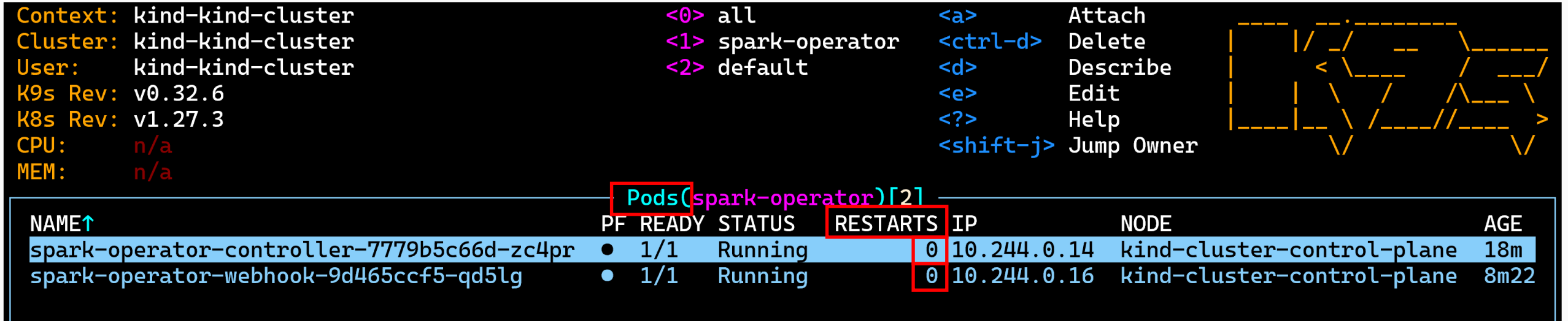
deployment after edit image version



1/1 means OK... all expected "../1" are running "1/.."

Before, it was error: "0/1"

Check : operator pods "Running"



Pods are "Running", and Ready "1/1"

Notice also Restart=0 (they are not in a CrashLoop causing many Restarts)

Finalize Installation:
create ServiceAccount + role
for your spark app

Adding serviceAccount "spark"

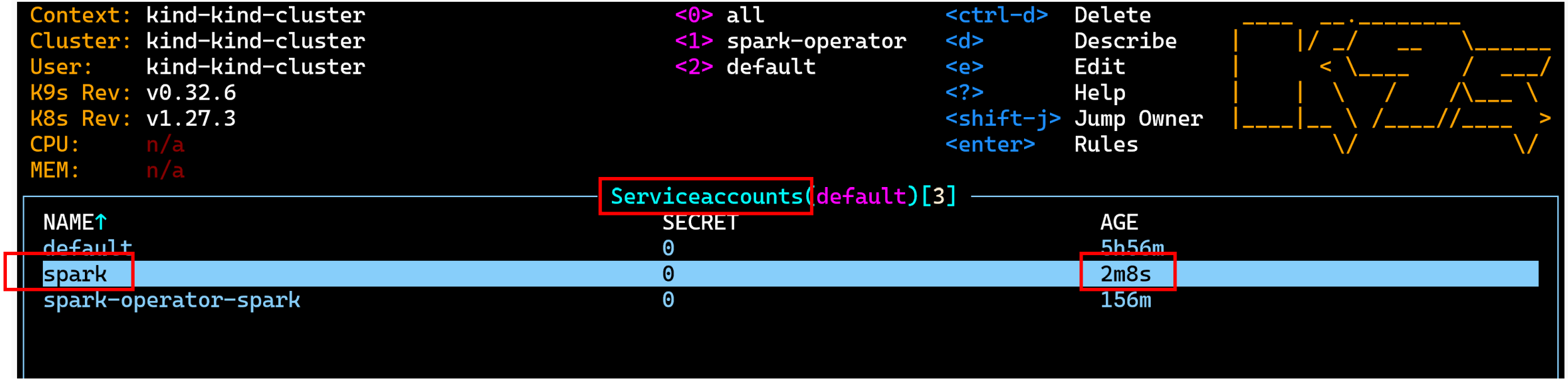
kubectl create serviceaccount spark

kubectl create clusterrolebinding spark-role --clusterrole=edit --serviceaccount=default:spark --namespace=default

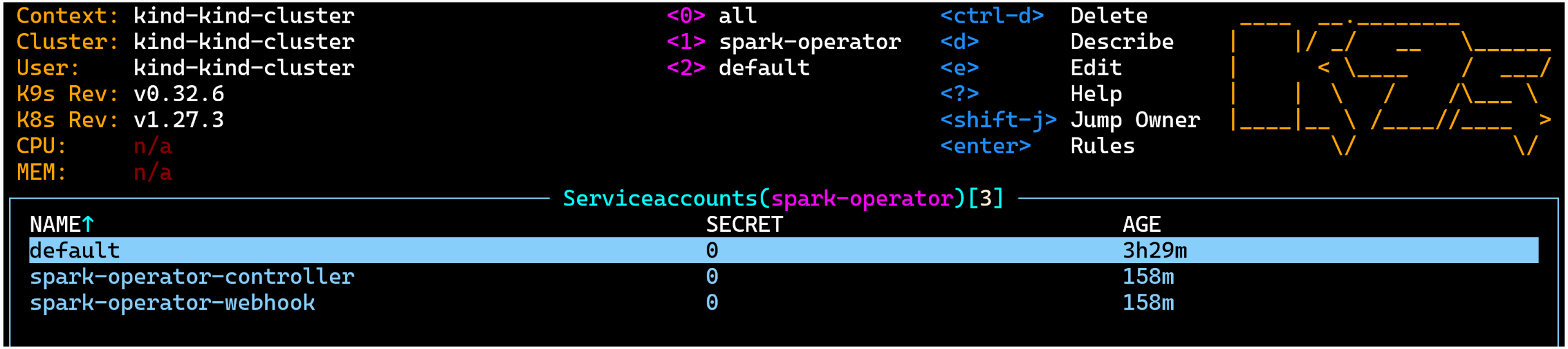
```
C:\apps\spark\spark-3.5.0> kubectl create serviceaccount spark
serviceaccount/spark created

C:\apps\spark\spark-3.5.0>kubectl create clusterrolebinding spark-role --clusterrole=edit --serviceaccount=default:spark --namespace=default
clusterrolebinding.rbac.authorization.k8s.io/spark-role created
```

Checking ServiceAccount



Notice: other serviceAccounts created by "Helm" on namespace "spark-operator"



Next Steps: Part 2/2

open document TD5-Spark-On-k8s-part2

part 2/2

(Optional) Launch spark-shell on Docker, in mode "--master local[*]"
(Optional) Launch spark with "spark-submit --master k8s://localhost:port"

Launch spark with "kubectl apply -f file.yaml"

(Optional... and backup if local install fails) Using Azure Kubernetes